Subject Name: Mechanics
Subject Code: ME1L001 (L-T-P-C: 3-1-0-4)

(L - Lecture; T - Tutorial; P - Practical; C – Credits)

Text Books:

- Vector Mechanics for Engineers: Statics and Dynamics - Ferdinand P. Beer, E. Russell Johnston, Jr. (TMH)
- Mechanics of Materials - Ferdinand Beer , E. Russell Johnston, Jr., J. DeWolf (TMH)

Reference Books:

- Engineering Mechanics: Statics and Dynamics - I.H. Shames (Pearson)
- Engineering Mechanics: Statics and Dynamics – J L Meriam and L G Kraige (Wiley)
- Engineering Mechanics - S. Timoshenko, D. H. Young (TMH)
- Materials of Materials – R C Hibbeler (Pearson)
- Elements of Strength of Materials - S. Timoshenko, D. H. Young (East West Press)
- Mechanics of Materials - James M. Gere, Barry J. Goodno (CL Engg)

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Course Contents

Force systems:

Moment of a force about a point and about an axis; couple moment; reduction of a force system to a force and a couple.

Equilibrium:

Free body diagram; equations of equilibrium; problems in two and three dimensions; plane frames and trusses.

Friction:

Laws of Coulomb friction, problems involving large and small contact surfaces; square threaded screws; belt friction; rolling resistance.

Kinematics and Kinetics of particles:

Particle dynamics in rectangular coordinates cylindrical coordinates and in terms of path variables; central force motion.

Properties of areas:

Moments of inertia and product of inertia of areas, polar moment of inertia, principal axes and principal moments of inertia.

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Course Contents

Concept of stress and strain:

Normal stress, shear stress, state of stress at a point, ultimate strength, allowable stress, factor of safety; normal strain, shear strain, Hooke's law, Poisson's ratio, generalized Hooke's law; analysis of axially loaded members.

Torsion:

Torsion of cylindrical bars, torsional stress, modulus of rigidity and deformation.

Flexural loading:

Shear and moment in beams; load, shear and moment relationship; shear and moment diagrams; flexure formula; shear stress in beams; differential equation of the elastic curve, deflection of beams.

Transformation of stress and strain:

Transformation of stress and strain, principal stresses, principal strains, Mohr's circle for stress and strain.

Combined loading:

Axial and torsional; axial and bending; axial, torsional and bending.

Column:

Buckling of slender columns, Euler bucking load for different end conditions

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What is Mechanics?

- Science- is the knowledge that comes from observing facts about universe carefully, carrying out experiments and making statements that are always true in particular conditions.
  Engineering- is the application of mathematics and science to the design and manufacture of items that benefit humanity.
  Design is the basic concept that distinguishes engineers from scientists.

- Mechanics may be defined as physical science which describes and predicts the conditions of rest or motion of bodies under the action of forces.

- Categories of Mechanics:
  - Rigid bodies
    - *Statics*
    - Dynamics
  - Deformable bodies
  - Fluids

- Mechanics is the foundation of most engineering sciences and is an indispensable prerequisite to their study.

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Fundamental Concepts

- *Continuum* –Pressure, density and temperature are actually the gross effects of the many molecules and atoms and they can be conveniently assumed to arise from a hypothetically continuous distribution of matter, which is called the continuum.

- *Particle* – defined as an object that has no size but that has a mass.

- *Rigid Bodies* – a combination of large number of particles occupying fixed positions with respect to each other.

- *Space* - associated with the notion of the position of a point P given in terms of three coordinates measured from a reference point or origin.
- *Time* - definition of an event requires specification of the time and position at which it occurred.
- *Mass* - used to characterize and compare bodies, e.g., response to earth's gravitational attraction and resistance to changes in translational motion.

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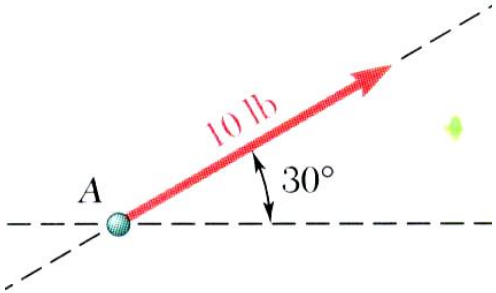
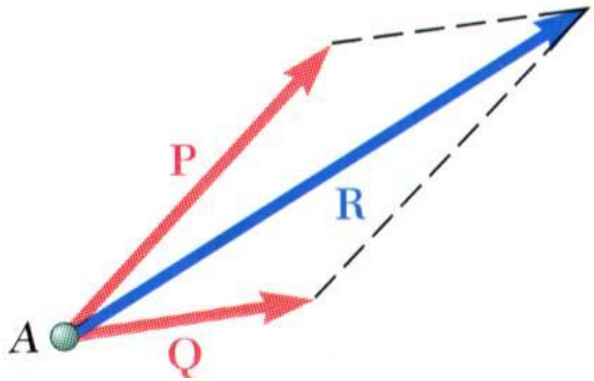
Resultant of Two Forces

- *Force* - represents the action of one body on another. A force is characterized by its point of application, magnitude, and direction, i.e., a force is a vector quantity.
- Experimental evidence shows that the combined effect of two forces may be represented by a single *resultant* force.





- The resultant is equivalent to the diagonal of a parallelogram which contains the two forces in adjacent legs.

In Newtonian Mechanics, space, time, and mass are absolute concepts, independent of each other. Force, however, is not independent of the other three. The force acting on a body is related to the mass of the body and the variation of its velocity with time.

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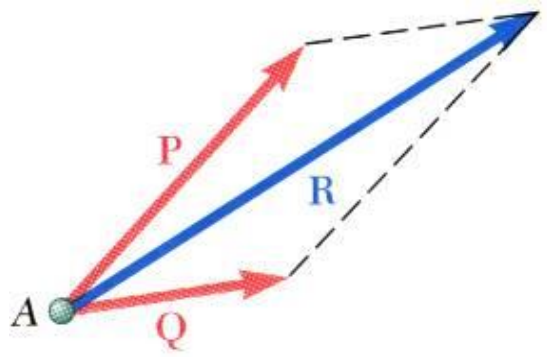
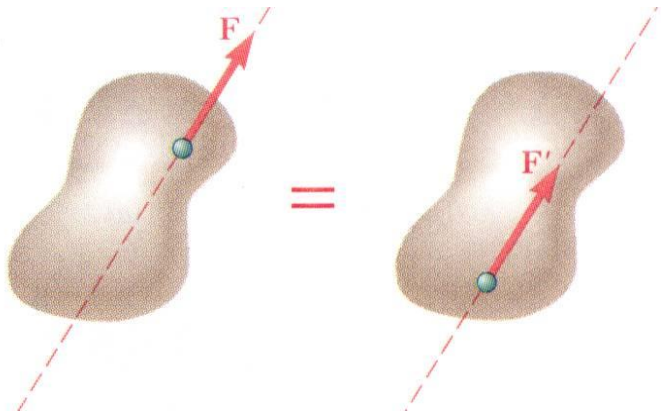
Fundamental Principles



- *Parallelogram Law*



- *Principle of Transmissibility*

- *Newton's First Law*: If the resultant force on a particle is zero, the particle will remain at rest or continue to move in a straight line.

- *Newton's Second Law*: A particle will have an acceleration proportional to a nonzero resultant applied force.

$$\vec{F} = m\vec{a}$$

- *Newton's Third Law*: The forces of action and reaction between two particles have the same magnitude and line of action with opposite sense.

- *Newton's Law of Gravitation*: Two particles are attracted with equal and opposite forces,

$$F = G\frac{Mm}{r^2} \qquad W = mg, \quad g = \frac{GM}{R^2}$$

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Systems of Units

- *Kinetic Units*: length, time, mass, and force.

- Three of the kinetic units, referred to as *basic units*, may be defined arbitrarily. The fourth unit, referred to as a *derived unit*, must have a definition compatible with Newton's 2nd Law,

$$\vec{F} = m\vec{a}$$

- *International System of Units* (SI): The basic units are length, time, and mass which are arbitrarily defined as the meter (m), second (s), and kilogram (kg). Force is the derived unit,

$$F = ma$$

$$1\,\mathrm{N} = (1\,\mathrm{kg})\left(1\,\frac{\mathrm{m}}{\mathrm{s}^2}\right)$$

- *U.S. Customary Units*: The basic units are length, time, and force which are arbitrarily defined as the foot (ft), second (s), and pound (lb). Mass is the derived unit,

$$m = \frac{F}{a}$$

$$1\,\mathrm{slug} = \frac{1\,\mathrm{lb}}{1\,\mathrm{ft/s}}$$

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Method of Problem Solution

- *Problem Statement*:
  Includes given data, specification of what is to be determined, and a figure showing all quantities involved.

- *Free-Body Diagrams*:
  Create separate diagrams for each of the bodies involved with a clear indication of all forces acting on each body.

- *Fundamental Principles*:
  The six fundamental principles are applied to express the conditions of rest or motion of each body. The rules of algebra are applied to solve the equations for the unknown quantities.

- *Solution Check*:
  - Test for errors in reasoning by verifying that the units of the computed results are correct,
  - test for errors in computation by substituting given data and computed results into previously unused equations based on the six principles,
  - **always** apply experience and physical intuition to assess whether results seem "reasonable"

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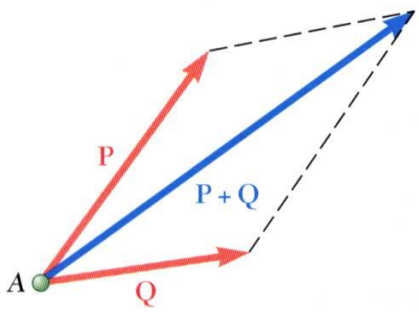
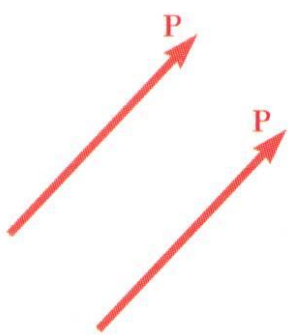
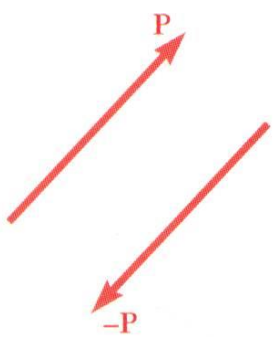
Vectors







- *Vector*: parameters possessing magnitude and direction which add according to the parallelogram law. Examples: displacements, velocities, accelerations.
- *Scalar*: parameters possessing magnitude but not direction. Examples: mass, volume, temperature
- Vector classifications:
  - *Fixed* or *bound* vectors have well defined points of application that cannot be changed without affecting an analysis.
  - *Free* vectors may be freely moved in space without changing their effect on an analysis.
  - *Sliding* vectors may be applied anywhere along their line of action without affecting an analysis.
- *Equal* vectors have the same magnitude and direction.
- *Negative* vector of a given vector has the same magnitude and the opposite direction.

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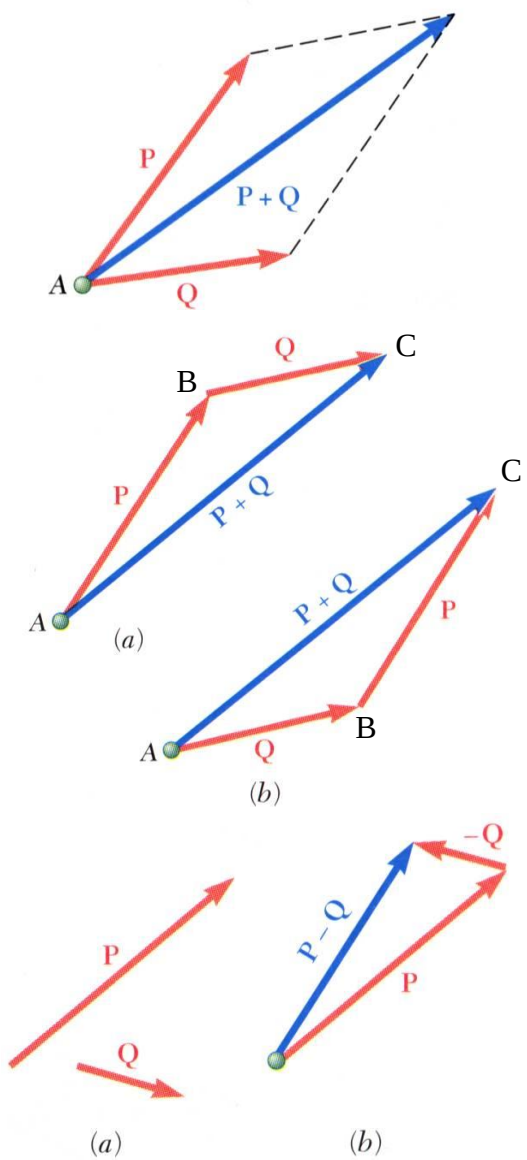
Addition of Vectors



- Trapezoid rule for vector addition
- Triangle rule for vector addition
- Law of cosines,

$$R^2 = P^2 + Q^2 - 2PQ\cos B$$
$$\vec{R} = \vec{P} + \vec{Q}$$

- Law of sines,

$$\frac{\sin A}{Q} = \frac{\sin B}{R} = \frac{\sin C}{A}$$

- Vector addition is commutative,

$$\vec{P} + \vec{Q} = \vec{Q} + \vec{P}$$

- Vector subtraction

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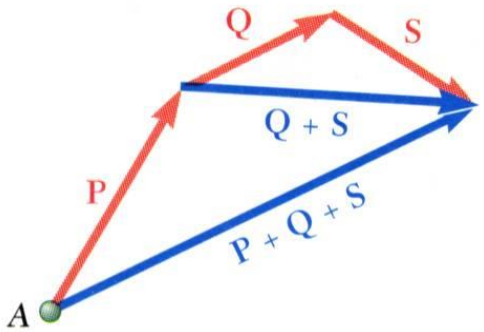
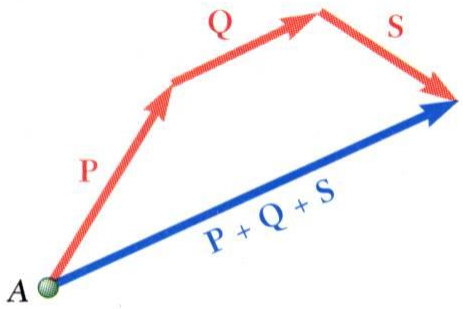
Addition of Vectors

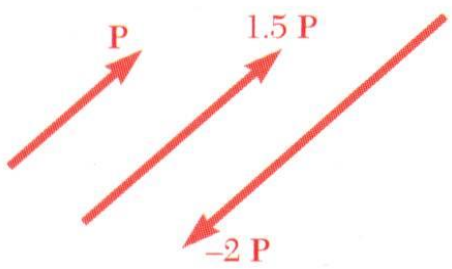


- Addition of three or more vectors through repeated application of the triangle rule



- The polygon rule for the addition of three or more vectors.
- Vector addition is associative,

$$\vec{P}+\vec{Q}+\vec{S}=\left(\vec{P}+\vec{Q}\right)+\vec{S}=\vec{P}+\left(\vec{Q}+\vec{S}\right)$$



- Multiplication of a vector by a scalar

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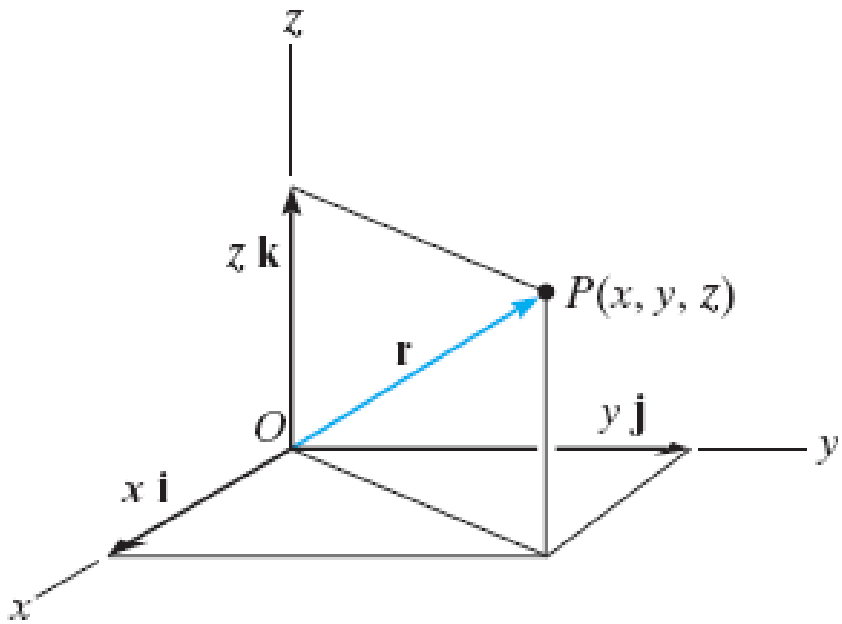
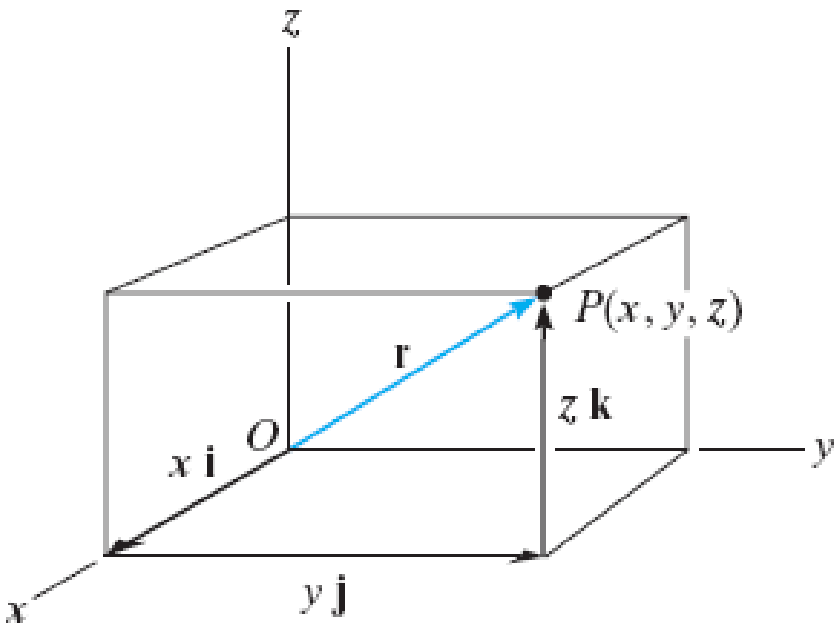
Position Vectors

Position Vector

- Position vector **r** is defined as a fixed vector which locates a point in space relative to another point.
- e.g. $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$





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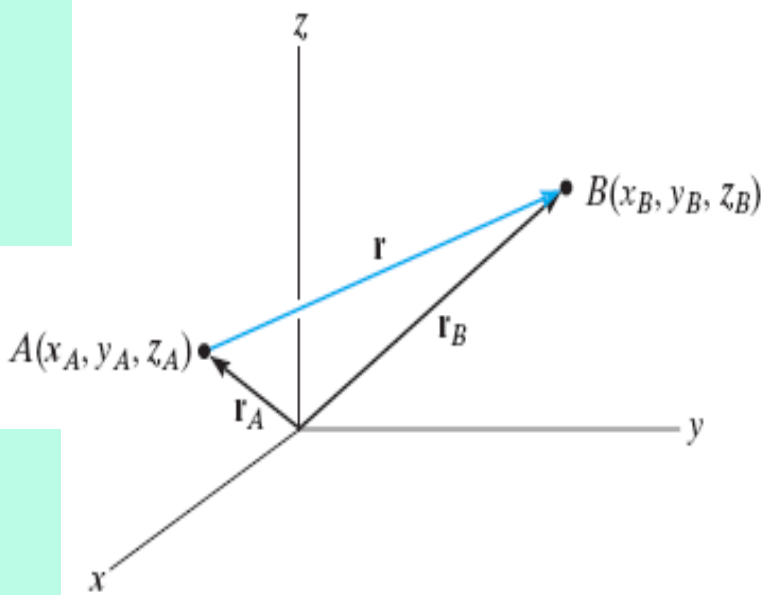
Position Vectors

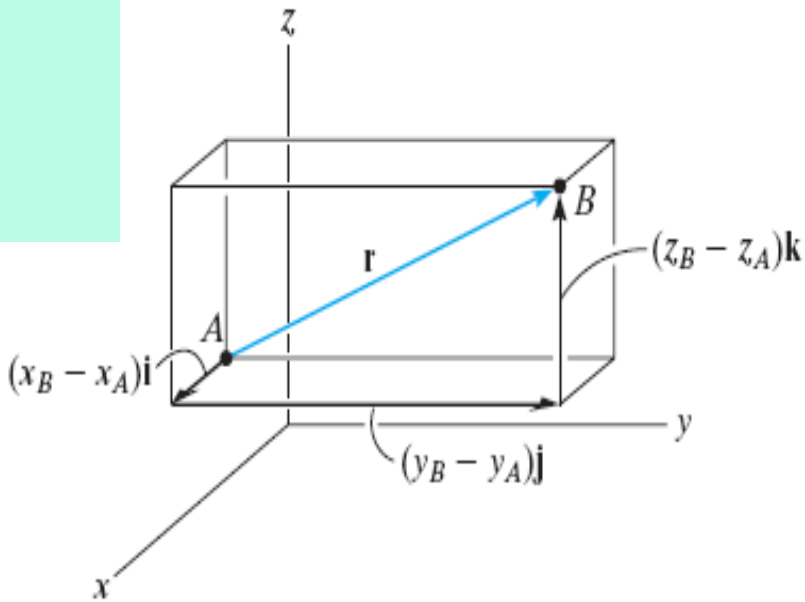
Position Vector

- Vector addition gives $\mathbf{r}_A + \mathbf{r} = \mathbf{r}_B$
- Solving

$$\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}$$

or $\mathbf{r} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}$





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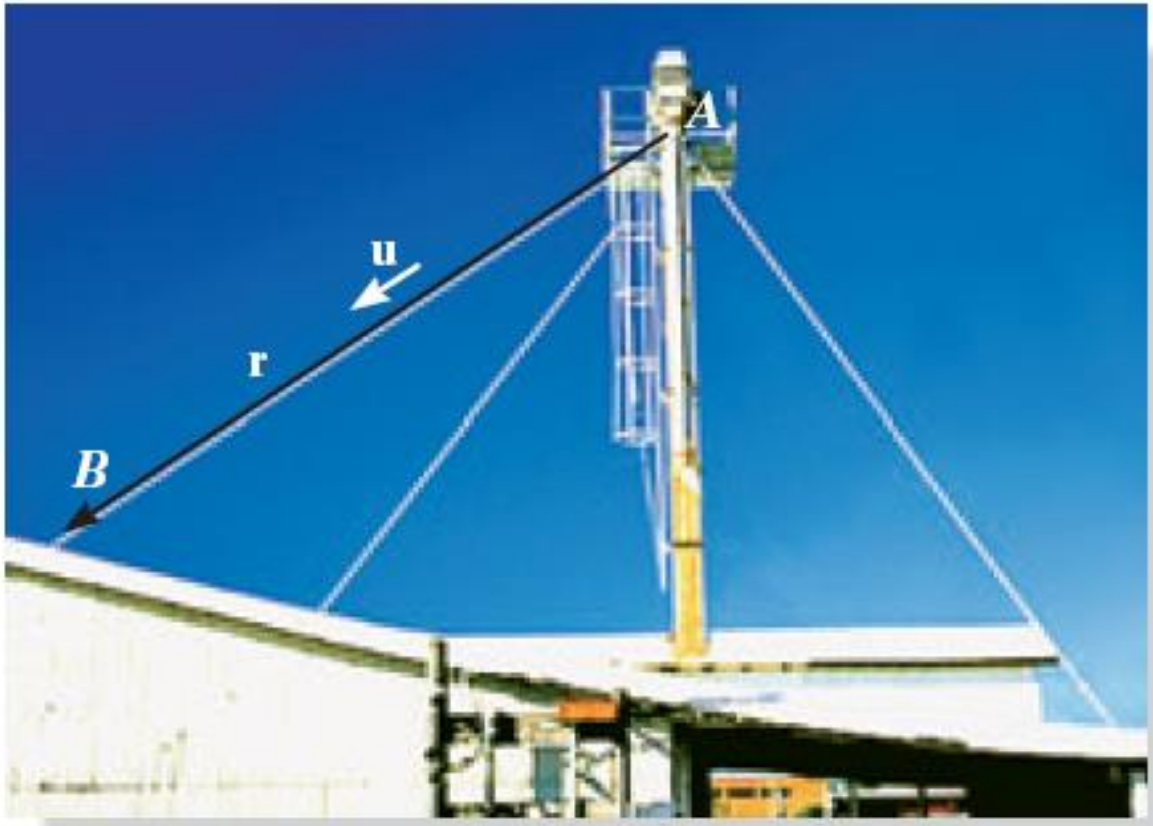
Position Vectors

- Length and direction of cable AB can be found by measuring A and B using the *x*, *y*, *z* axes
- Position vector **r** can be established
- Magnitude r represent the length of cable
- Angles, α, β and γ represent the direction of the cable
- Unit vector, **u** = **r**/r



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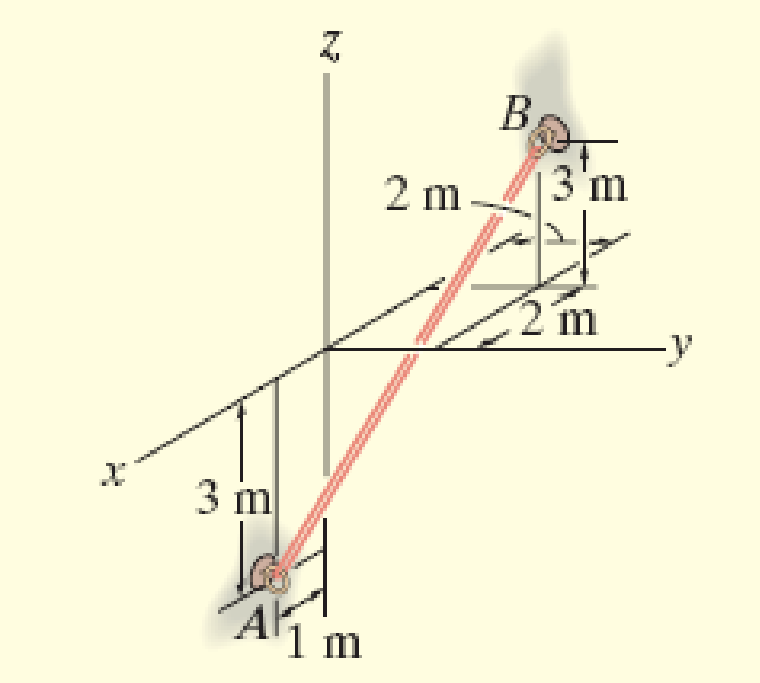
Example

An elastic rubber band is attached to points A and B. Determine its length and its direction measured from A towards B.



Solution:

Position vector

r = [-2m – 1m]i + [2m – 0]j + [3m – (-3m)]k

= {-3i + 2j + 6k}m

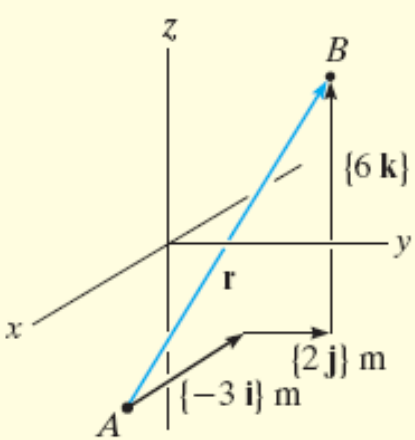
Magnitude = length of the rubber band

$$r = \sqrt{(-3)^2 + (2)^2 + (6)^2} = 7m$$

Unit vector in the director of **r**

u = **r** /*r*

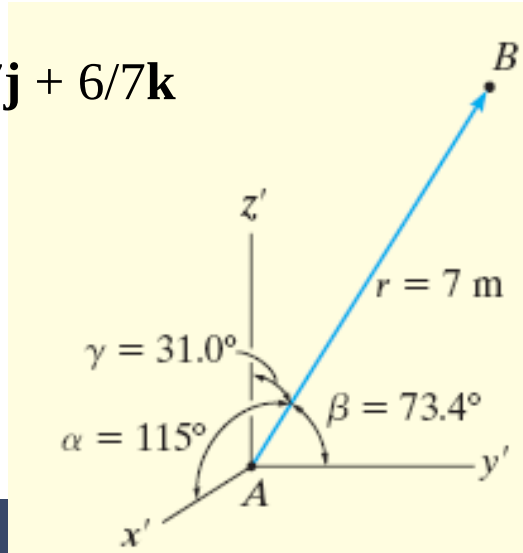
= -3/7**i** + 2/7**j** + 6/7**k**



$\alpha = \cos^{-1}(-3/7) = 115^\circ$

$\beta = \cos^{-1}(2/7) = 73.4^\circ$

$\gamma = \cos^{-1}(6/7) = 31.0^\circ$



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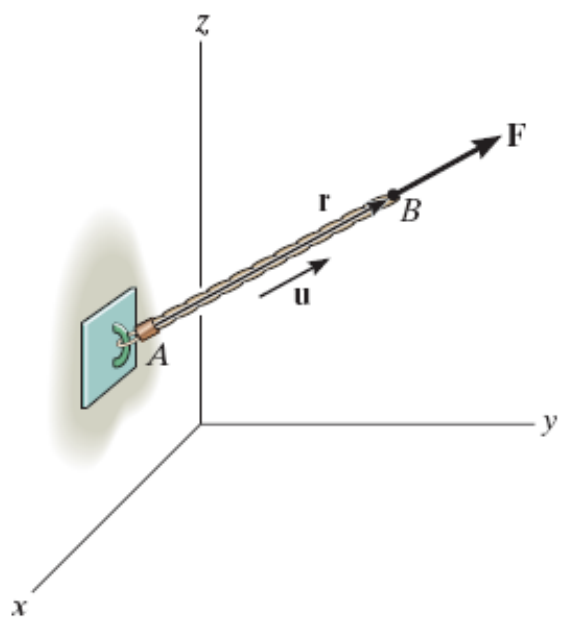
Force Vector Directed along a Line

- In 3D problems, direction of **F** is specified by 2 points, through which its line of action lies
- **F** can be formulated as a Cartesian vector

$$\mathbf{F} = F\,\mathbf{u} = F\,(\mathbf{r}/r)$$

- Note that **F** has units of forces (N) unlike **r**, with units of length (m)

- Force **F** acting along the chain can be presented as a Cartesian vector by
  - Establish *x*, *y*, *z* axes
  - Form a position vector **r** along length of chain
- Unit vector, $\mathbf{u} = \mathbf{r}/r$ that defines the direction of both the chain and the force
- We get $\mathbf{F} = F\mathbf{u}$



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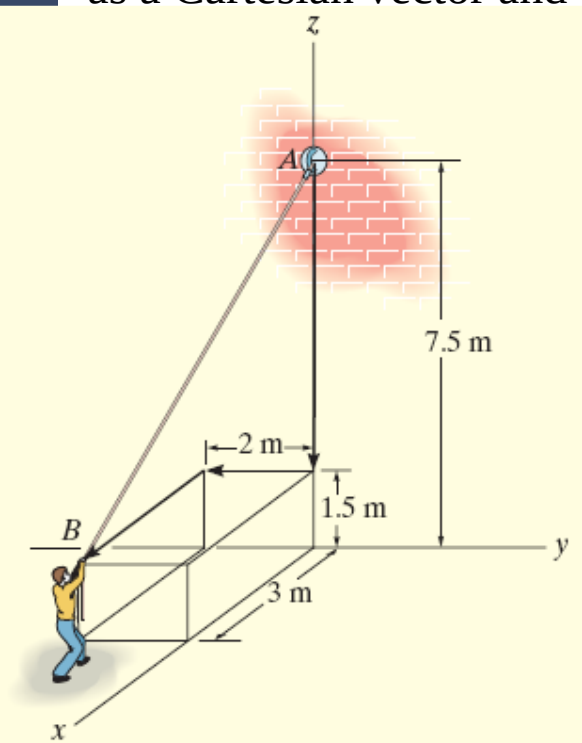
Example

The man pulls on the cord with a force of 350N. Represent this force acting on the support A, as a Cartesian vector and determine its direction.



Solution:

End points of the cord are A (0m, 0m, 7.5m) and B (3m, -2m, 1.5m)

$\mathbf{r}$ = (3m − 0m)**i** + (-2m − 0m)**j** + (1.5m − 7.5m)**k**

= {3**i** − 2**j** − 6**k**}m

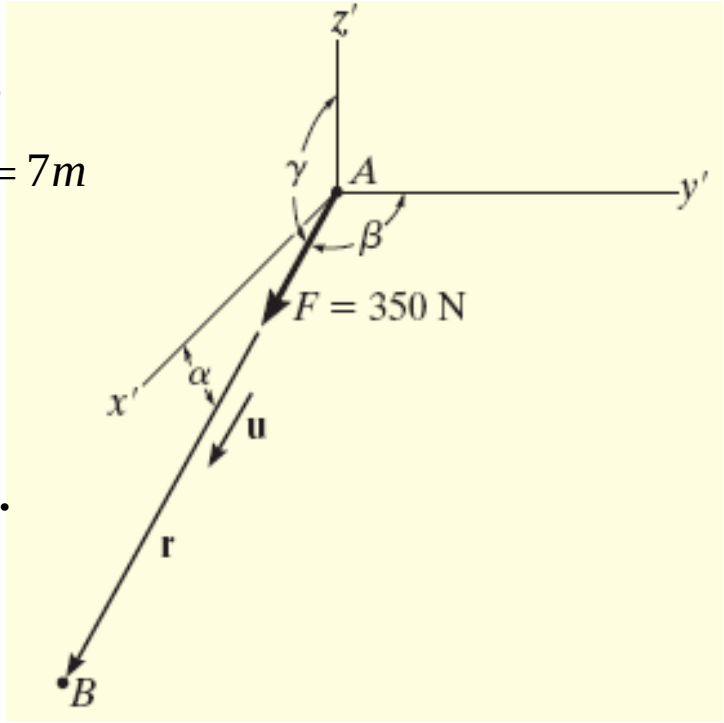
Magnitude = length of cord AB

$$r = \sqrt{(3m)^2 + (-2m)^2 + (-6m)^2} = 7m$$

Unit vector,

u = **r** /*r*

= 3/7**i** - 2/7**j** - 6/7**k**



Force **F** has a magnitude of 350N, direction specified by **u.**

F = *F***u**

= 350N(3/7**i** - 2/7**j** - 6/7**k**)

= {150**i** - 100**j** - 300**k**} N

$\alpha = \cos^{-1}(3/7) = 64.6^\circ$, $\beta = \cos^{-1}(-2/7) = 107^\circ$, $\gamma = \cos^{-1}(-6/7) = 149^\circ$

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Rigid Bodies: Equivalent Systems of Forces

- Treatment of a body as a single particle is not always possible. In general, the size of the body and the specific points of application of the forces must be considered.
- Most bodies in elementary mechanics are assumed to be rigid, i.e., the actual deformations are small and do not affect the conditions of equilibrium or motion of the body.
- The present discussion describes the effect of forces exerted on a rigid body and how to replace a given system of forces with a simpler equivalent system.
  - moment of a force about a point
  - moment of a force about an axis
  - moment due to a couple
- Any system of forces acting on a rigid body can be replaced by an equivalent system consisting of one force acting at a given point and one couple.

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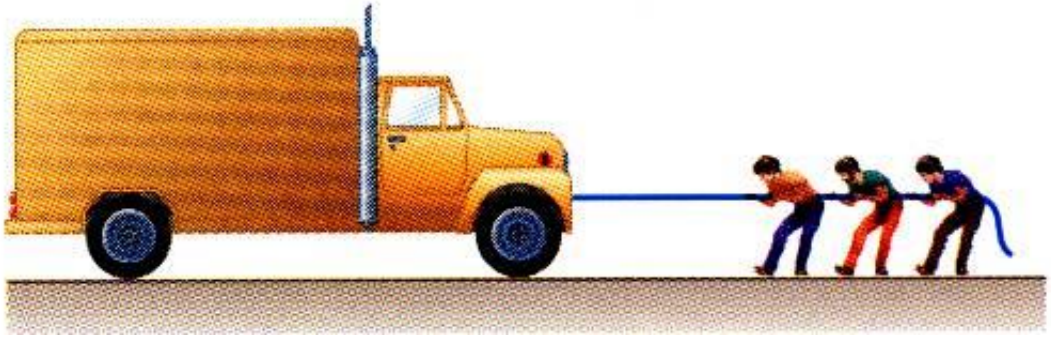
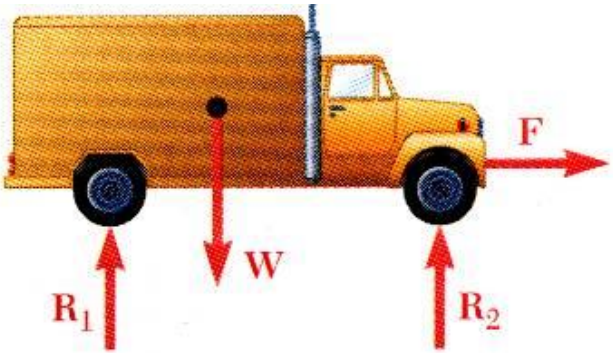
External and Internal Forces

- Forces acting on rigid bodies are divided into two groups:
  - External forces
  - Internal forces



- External forces are shown in a free-body diagram.



- If unopposed, each external force can impart a motion of translation or rotation, or both.

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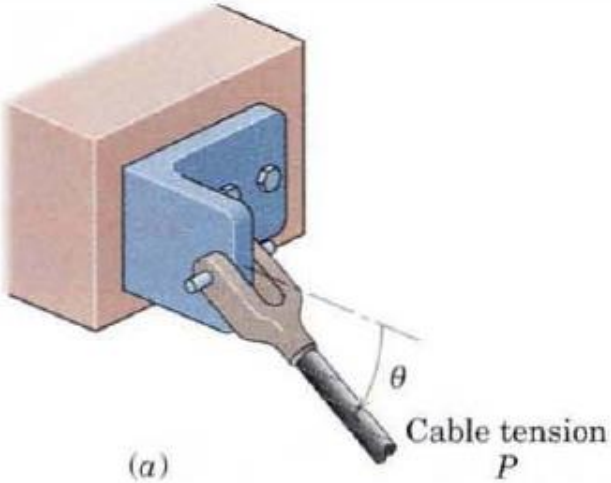
External forces represent the action of other bodies on the rigid body under consideration. They entirely responsible for the external behavior of the rigid body. They will either cause it to move or ensure that it remains at rest.

Internal forces are the forces which hold together the particles forming the rigid body. If the rigid body is structurally composed of several parts, the forces holding the components together are also defined as internal forces.

In other words, the action of a force can be separated into two effects:

The effects of P external to the bracket are reactive forces (not shown) exerted on the bracket by the foundation and bolts because of the action of P. Forces external to a body can be either applied forces or reactive forces.

The effects of P internal to the bracket are the resulting internal forces and deformations distributed throughout the material of the bracket.



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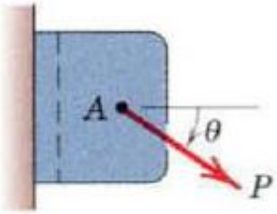

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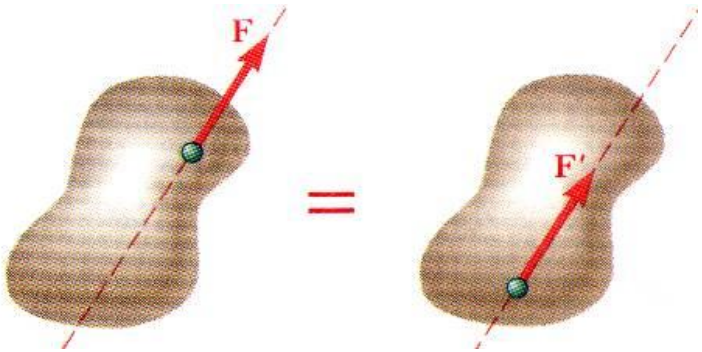
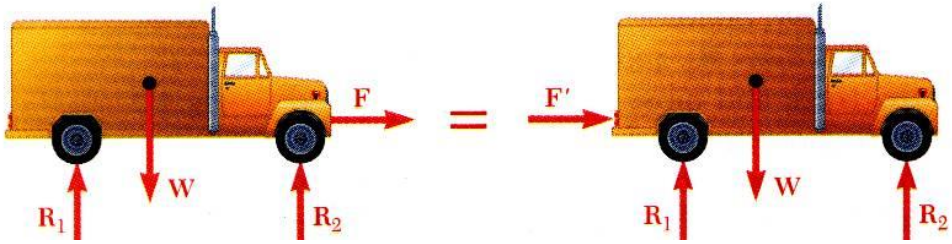
Principle of Transmissibility: Equivalent Forces

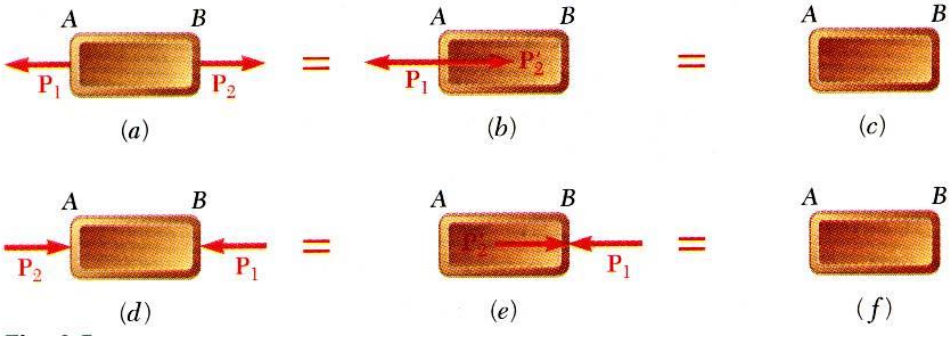
- *Principle of Transmissibility* - Conditions of equilibrium or motion are not affected by *transmitting* a force along its line of action.
  NOTE: **F** and **F**' are equivalent forces.



- Moving the point of application of the force **F** to the rear bumper does not affect the motion or the other forces acting on the truck.



- Principle of transmissibility may not always apply in determining internal forces and deformations.



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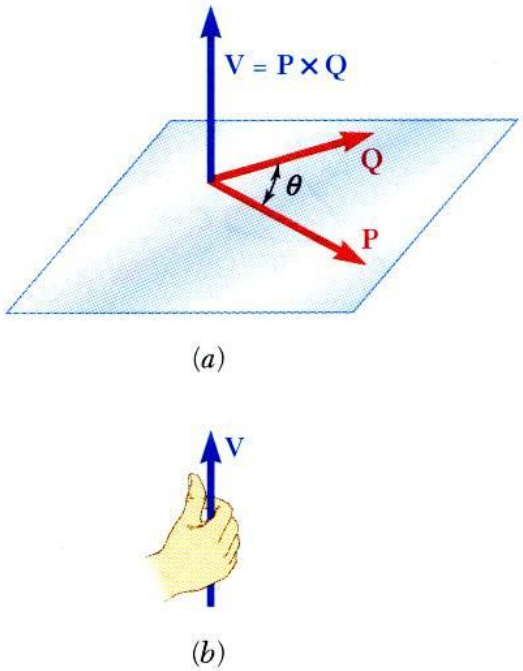
Vector Product of Two Vectors

- Concept of the moment of a force about a point is more easily understood through applications of the *vector product* or *cross product*.

- Vector product of two vectors $\boldsymbol{P}$ and $\boldsymbol{Q}$ is defined as the vector $\boldsymbol{V}$ which satisfies the following conditions:
  1. Line of action of $\boldsymbol{V}$ is perpendicular to plane containing $\boldsymbol{P}$ and $\boldsymbol{Q}$.
  2. Magnitude of $\boldsymbol{V}$ is $V = PQ\sin\theta$
  3. Direction of $\boldsymbol{V}$ is obtained from the right-hand rule.



- Vector products:
  - are not commutative, $\boldsymbol{Q}\times\boldsymbol{P} = -(\boldsymbol{P}\times\boldsymbol{Q})$
  - are distributive, $\boldsymbol{P}\times(\boldsymbol{Q}_1+\boldsymbol{Q}_2) = \boldsymbol{P}\times\boldsymbol{Q}_1 + \boldsymbol{P}\times\boldsymbol{Q}_2$
  - are not associative, $(\boldsymbol{P}\times\boldsymbol{Q})\times\boldsymbol{S} \neq \boldsymbol{P}\times(\boldsymbol{Q}\times\boldsymbol{S})$

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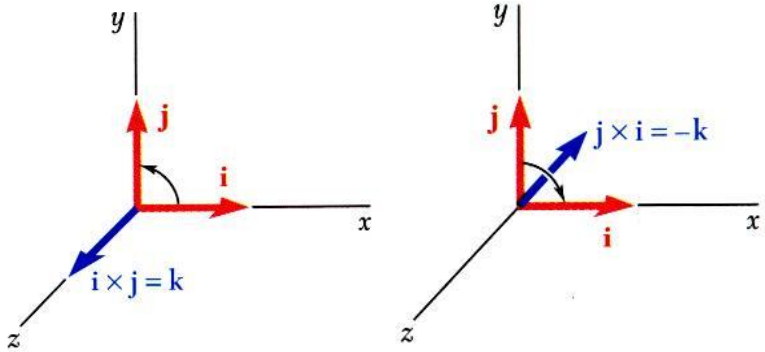
Vector Products: Rectangular Components

- Vector products of Cartesian unit vectors,

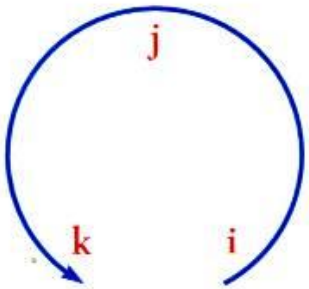
$$\vec{i}\times\vec{i}=0 \qquad \vec{j}\times\vec{i}=-\vec{k} \qquad \vec{k}\times\vec{i}=\vec{j}$$
$$\vec{i}\times\vec{j}=\vec{k} \qquad \vec{j}\times\vec{j}=0 \qquad \vec{k}\times\vec{j}=-\vec{i}$$
$$\vec{i}\times\vec{k}=-\vec{j} \qquad \vec{j}\times\vec{k}=\vec{i} \qquad \vec{k}\times\vec{k}=0$$



- Vector products in terms of rectangular coordinates

$$\vec{V}=\left(P_x\vec{i}+P_y\vec{j}+P_z\vec{k}\right)\times\left(Q_x\vec{i}+Q_y\vec{j}+Q_z\vec{k}\right)$$
$$=\left(P_yQ_z-P_zQ_y\right)\vec{i}+\left(P_zQ_x-P_xQ_z\right)\vec{j}$$
$$+\left(P_xQ_y-P_yQ_x\right)\vec{k}$$

$$=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ P_x & P_y & P_z\\ Q_x & Q_y & Q_z\end{vmatrix}$$



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Moment of a Force About a Point

- A force vector is defined by its magnitude and direction. Its effect on the rigid body also depends on it point of application.

- The *moment* of $\boldsymbol{F}$ about O is defined as

$$\boldsymbol{M_O} = \boldsymbol{r} \times \boldsymbol{F}$$

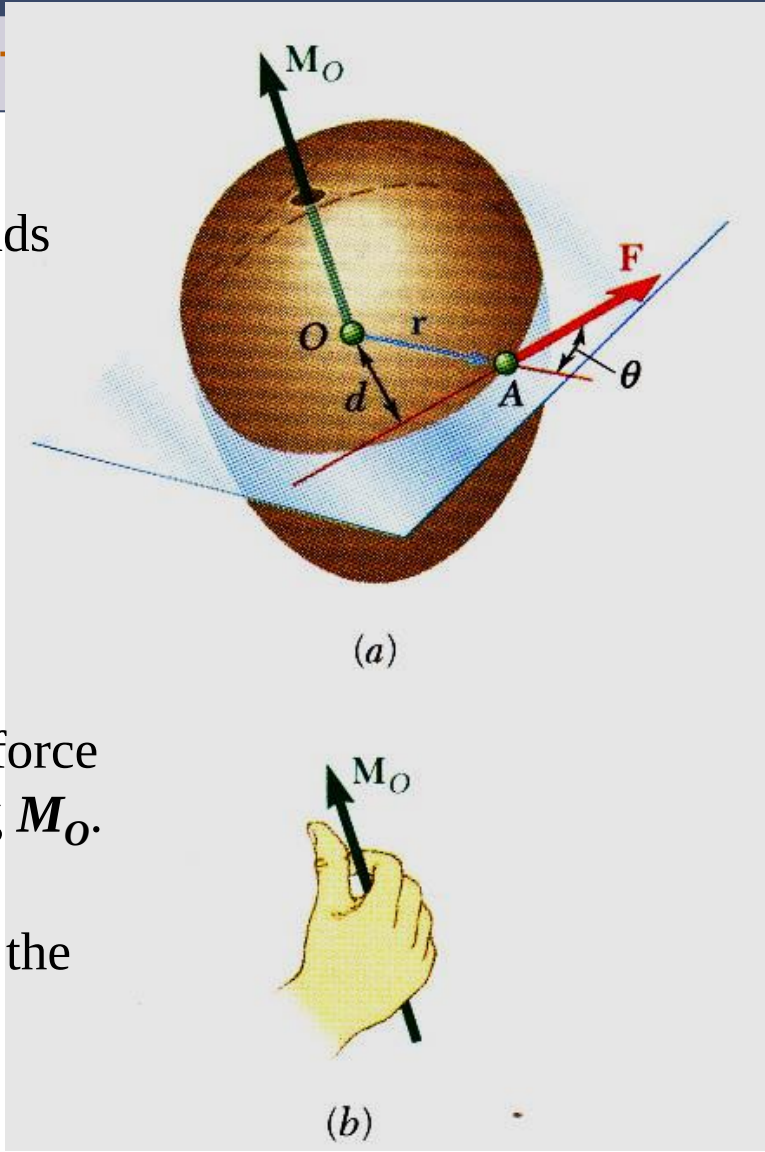
- The moment vector $\boldsymbol{M_O}$ is perpendicular to the plane containing O and the force $\boldsymbol{F}$.

- Magnitude of $\boldsymbol{M_O}$ measures the tendency of the force to cause rotation of the body about an axis along $\boldsymbol{M_O}$.

$$M_O = rF\sin\theta = Fd$$

The sense of the moment may be determined by the right-hand rule.

- Any force $\boldsymbol{F'}$ that has the same magnitude and direction as $\boldsymbol{F}$, is *equivalent* if it also has the same line of action and therefore, produces the same moment.



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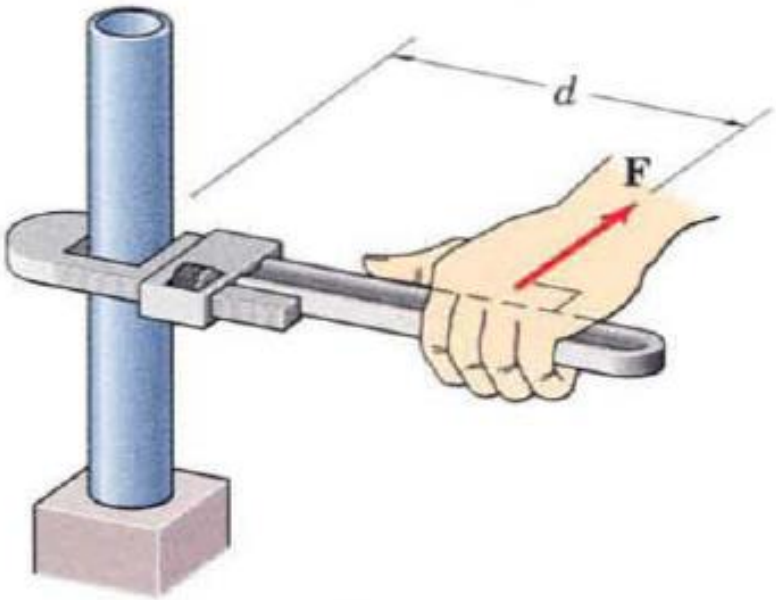
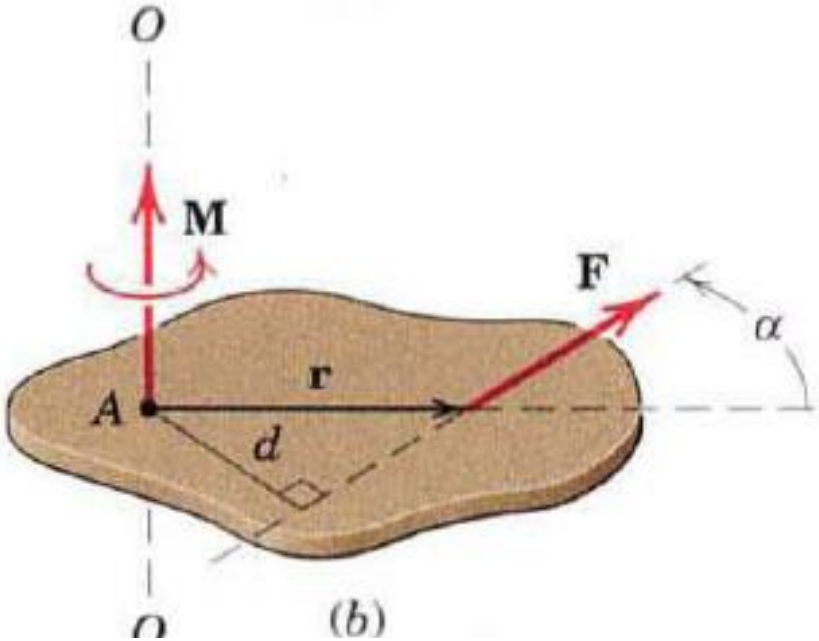
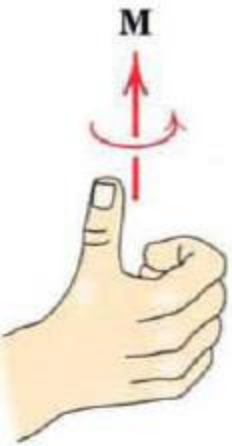
Moment of a Force About a Point



(a)



(b)



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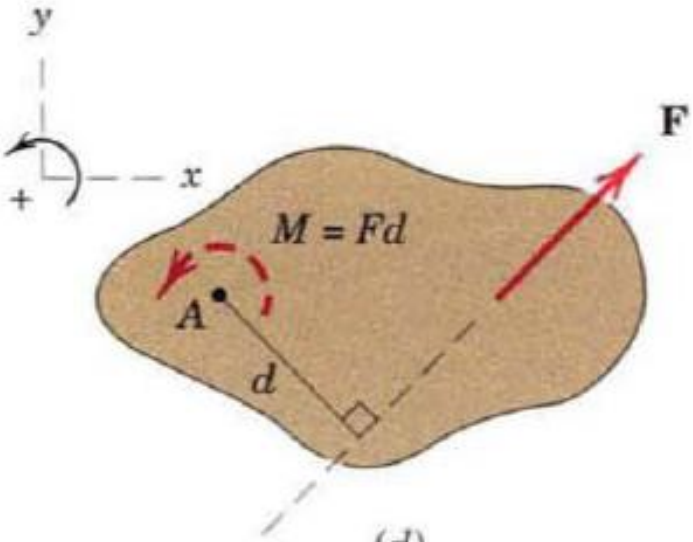

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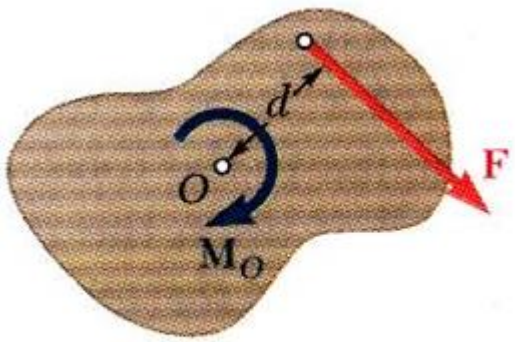
Moment of a Force About a Point

- *Two-dimensional structures* have length and breadth but negligible depth and are subjected to forces contained in the plane of the structure.

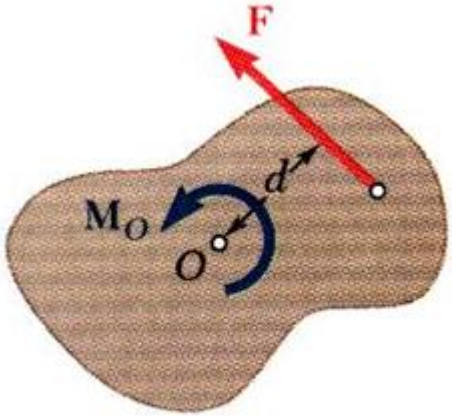
- The plane of the structure contains the point *O* and the force $\boldsymbol{F}$. $\boldsymbol{M_O}$, the moment of the force about *O* is perpendicular to the plane.

- If the force tends to rotate the structure clockwise, the sense of the moment vector is out of the plane of the structure and the magnitude of the moment is positive.

- If the force tends to rotate the structure counterclockwise, the sense of the moment vector is into the plane of the structure and the magnitude of the moment is negative.



(*a*) $\boldsymbol{M}_O = \quad Fd$



(*b*) $\boldsymbol{M}_O = -Fd$

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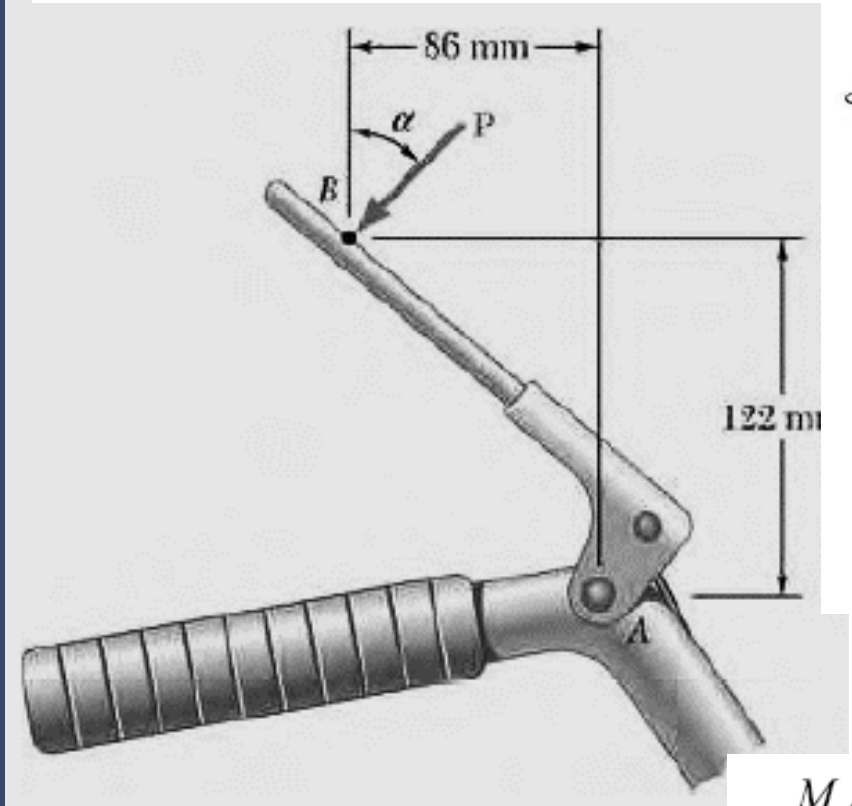
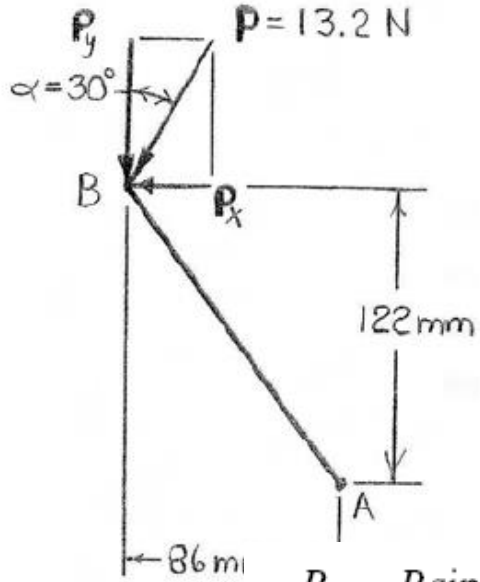
Example Problem

A 13.2-N force **P** is applied to the lever which controls the auger of a snowblower. Determine the moment of **P** about A when α is equal to 30°.





$$P_x = P\sin\alpha = (13.2\text{ N})\sin 30° = 6.60\text{ N}$$

$$P_y = P\cos\alpha = (13.2\text{ N})\cos 30° = 11.4315\text{ N}$$

$$M_A = x_{B/A}P_y + y_{B/A}P_x$$

$$= (0.086\text{ m})(11.4315\text{ N}) + (0.122\text{ m})(6.60\text{ N})$$

$$= 1.78831\text{ N}\cdot\text{m}$$

or $\mathbf{M}_A = 1.788\text{ N}\cdot\text{m}$ ↻

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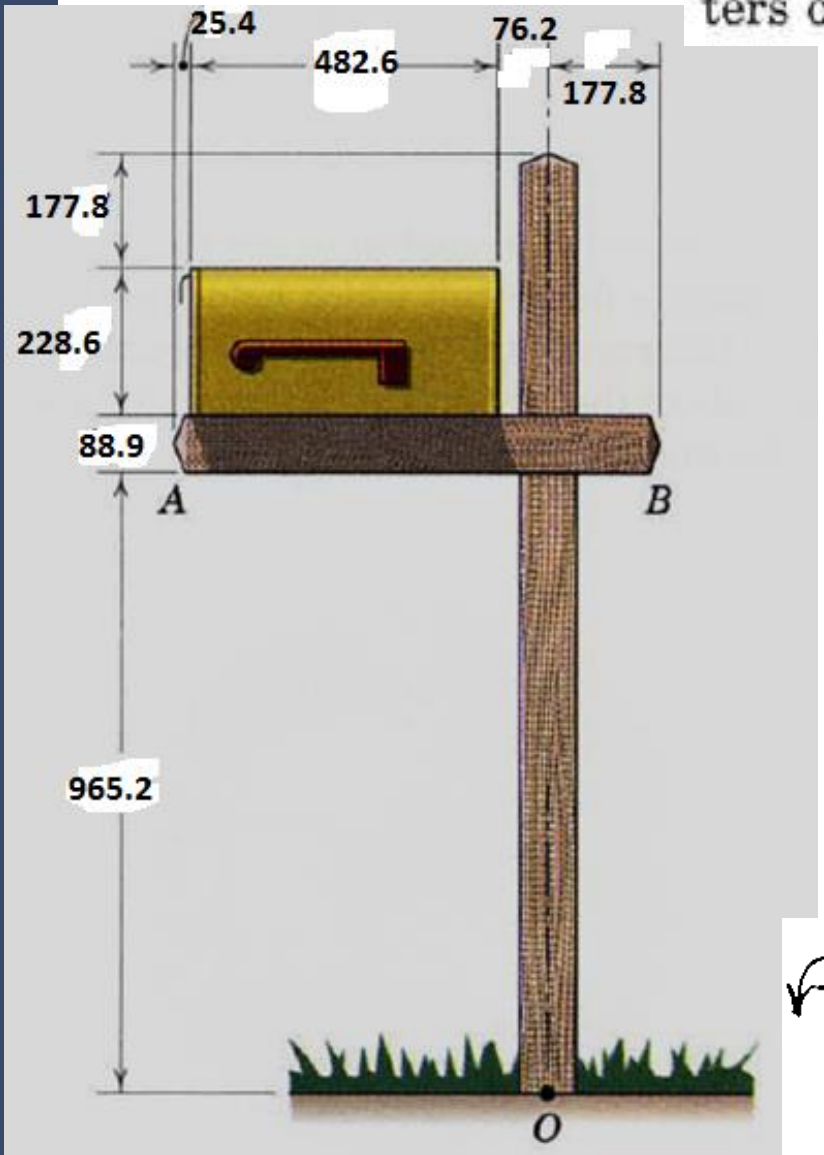
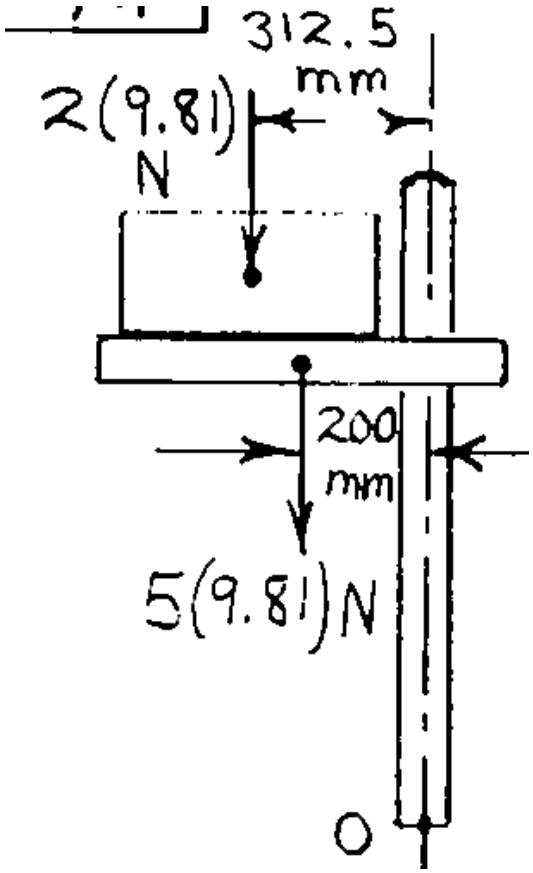
Sample Problem

Determine the combined moment about O due to the weight of the mailbox and the cross member AB. The mailbox weighs 4 lb and the uniform cross member weighs 10 lb. Both weights act at the geometric centers of the respective items.

1 lb ≈ 0.5 kg





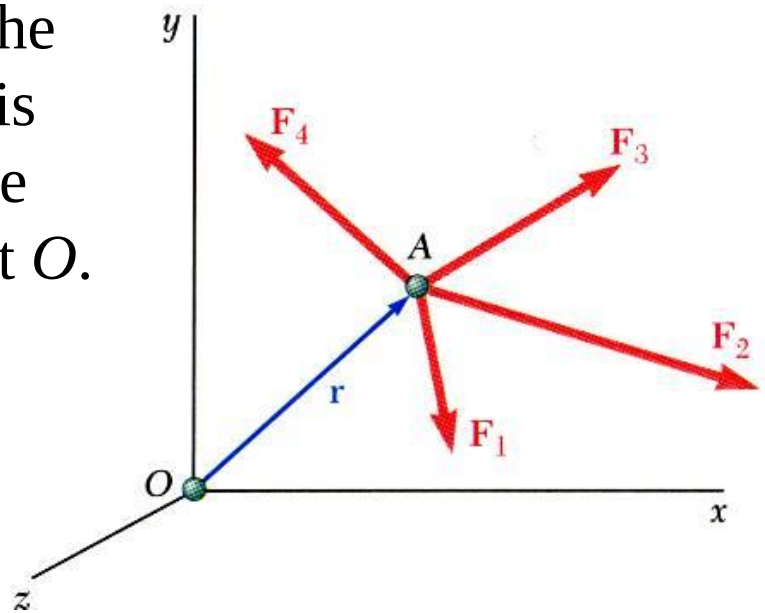
$$\circlearrowleft + \quad M_O = 2(9.81)(0.3125) + 5(9.81)(0.2)$$

$$= 15.94 \text{ N}\cdot\text{m} \quad \text{CCW}$$

Varignon's Theorem

- The moment about a give point *O* of the resultant of several concurrent forces is equal to the sum of the moments of the various moments about the same point *O*.

$$\vec{r} \times \left(\vec{F}_1 + \vec{F}_2 + \cdots\right) = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2 + \cdots$$



- Varigon's Theorem makes it possible to replace the direct determination of the moment of a force ***F*** by the moments of two or more component forces of ***F***.

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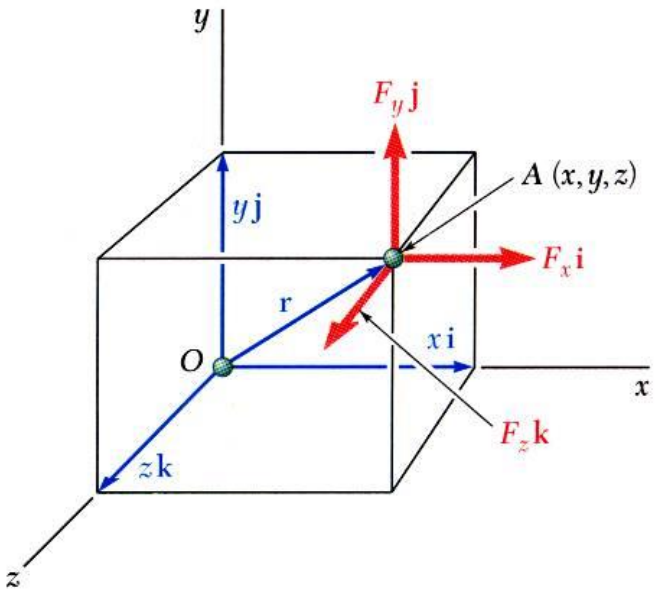
Rectangular Components of the Moment of a Force

The moment of ***F*** about *O*,

$$\vec{M}_O = \vec{r} \times \vec{F}, \quad \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$



$$\vec{M}_O = M_x\vec{i} + M_y\vec{j} + M_z\vec{k}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \left(yF_z - zF_y\right)\vec{i} + \left(zF_x - xF_z\right)\vec{j} + \left(xF_y - yF_x\right)\vec{k}$$

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Rectangular Components of the Moment of a Force

The moment of ***F*** about *B*,

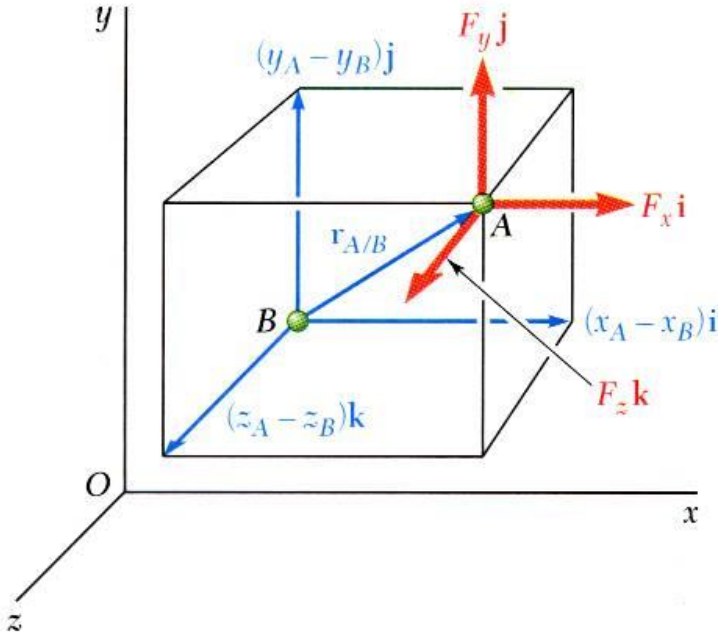
$$\vec{M}_B = \vec{r}_{A/B} \times \vec{F}$$

$$\vec{r}_{A/B} = \vec{r}_A - \vec{r}_B$$

$$= (x_A - x_B)\vec{i} + (y_A - y_B)\vec{j} + (z_A - z_B)\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\vec{M}_B = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ (x_A - x_B) & (y_A - y_B) & (z_A - z_B) \\ F_x & F_y & F_z \end{vmatrix}$$



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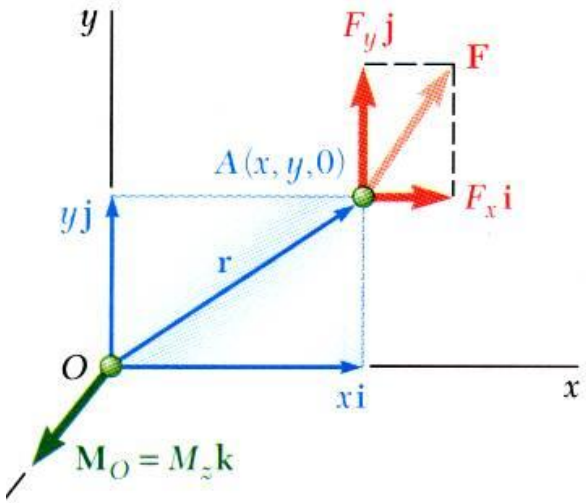
Rectangular Components of the Moment of a Force

For two-dimensional structures,

$$\vec{M}_O = \left(xF_y - yF_z\right)\vec{k}$$

$$M_O = M_Z$$

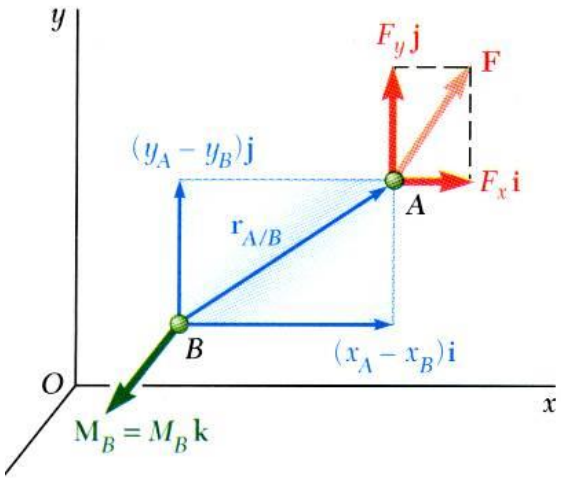
$$= xF_y - yF_z$$



$$\vec{M}_O = \left[(x_A - x_B)F_y - (y_A - y_B)F_z\right]\vec{k}$$

$$M_O = M_Z$$

$$= (x_A - x_B)F_y - (y_A - y_B)F_z$$



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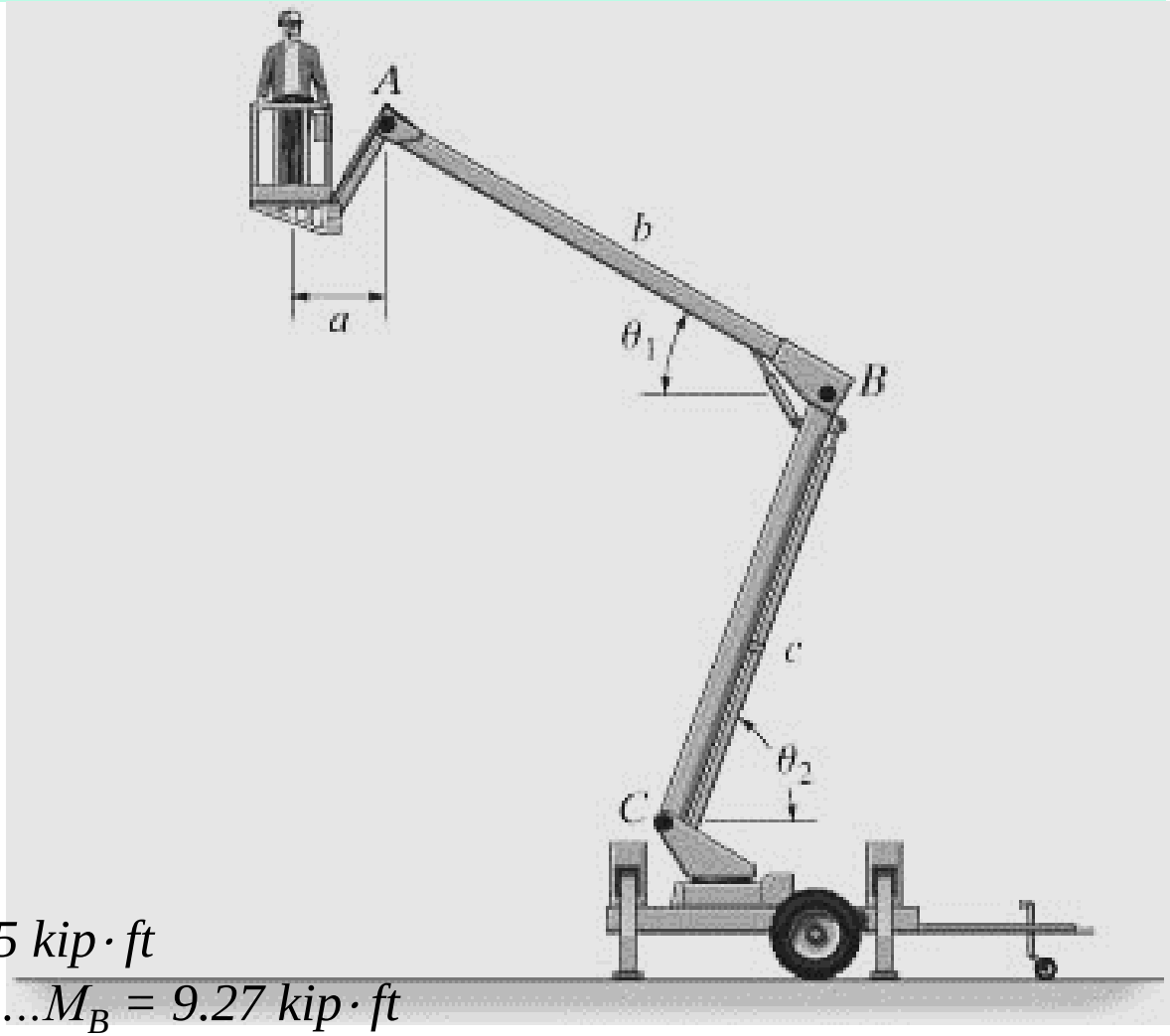
The Snorkel Co.produces the articulating boom platform that can support weight *W. If the* boom is in the position shown, determine the moment of this force about points *A, B, and C.*

kip = 10^3lb



Given:

$a = 3\ ft$

$b = 16\ ft$

$c = 15\ ft$

$\theta_1 = 30\ deg$

$\theta_2 = 70\ deg$

$W = 550\ lb$

Solution:

$M_A = Wa$…….. $M_A = 1.65\ kip \cdot ft$

$M_B = W(a + b\cos(\theta_1))$ ……..$M_B = 9.27\ kip \cdot ft$

$M_C = W(a + b\cos(\theta_1) - c\cos(\theta_2))$..... $M_C = 6.45\ kip \cdot ft$

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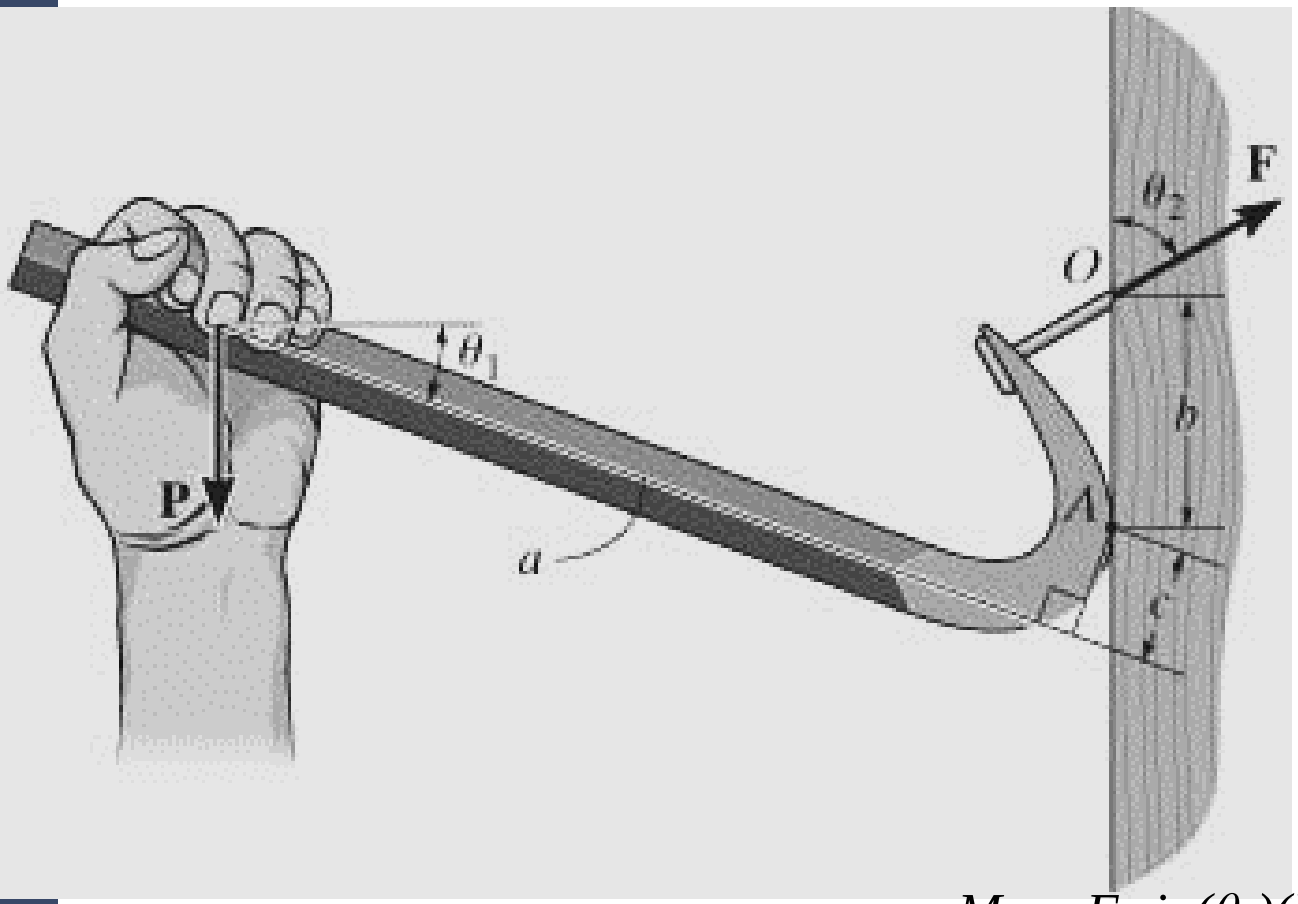
If it takes a force F to pull the nail out, determine the smallest vertical force P that must be applied to the handle of the crowbar.



Given:

$F = 125\ lb$

$a = 14\ in$

$b = 3\ in$

$c = 1.5\ in$

$\theta_1 = 20\ deg$

$\theta_2 = 60\ deg$

$M_F = F \sin(\theta_2)(b) \ldots\ldots\ldots\ldots M_F = 325\ lb \cdot in$

$P\ a \cos\theta_1 + c \sin\theta_1 = M_F$

$P = M_F/[(a)\cos(\theta_1) + (c)\sin(\theta_1)]$

$P = 23.8\ lb$

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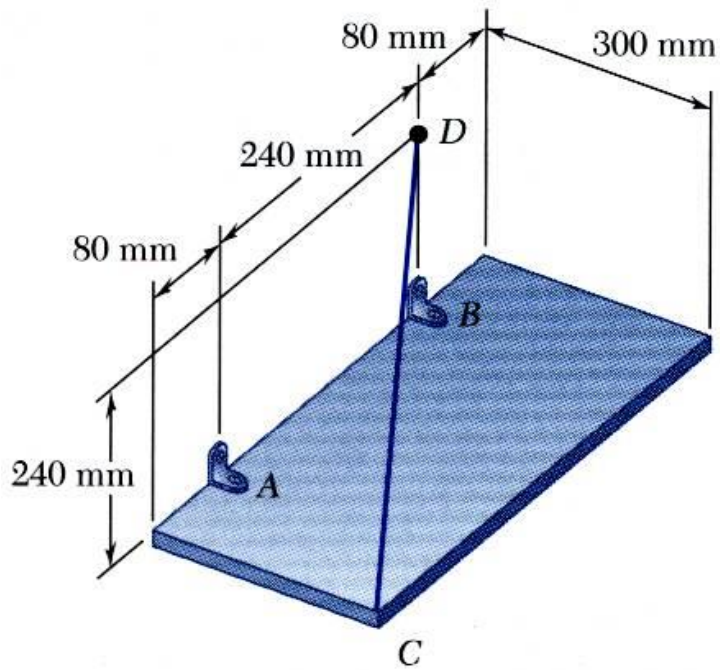
Sample Problem 3.4



The rectangular plate is supported by the brackets at *A* and *B* and by a wire *CD*. Knowing that the tension in the wire is 200 N, determine the moment about *A* of the force exerted by the wire at *C*.

SOLUTION:

The moment M_A of the force F exerted by the wire is obtained by evaluating the vector product,

$$\vec{M}_A = \vec{r}_{C/A} \times \vec{F}$$

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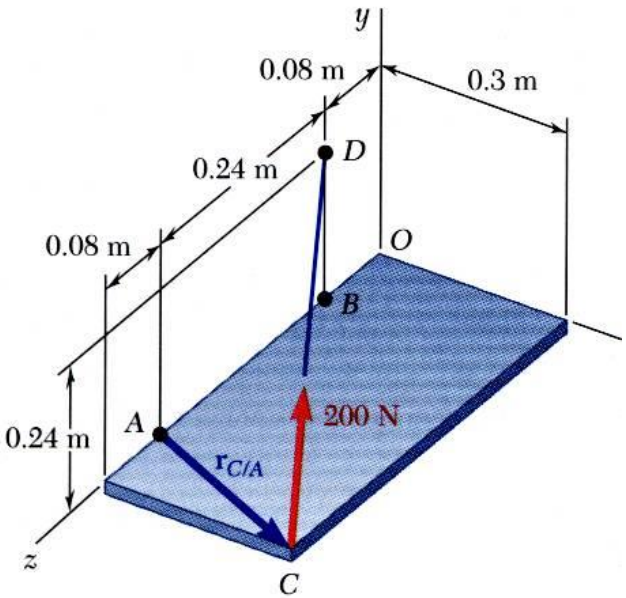
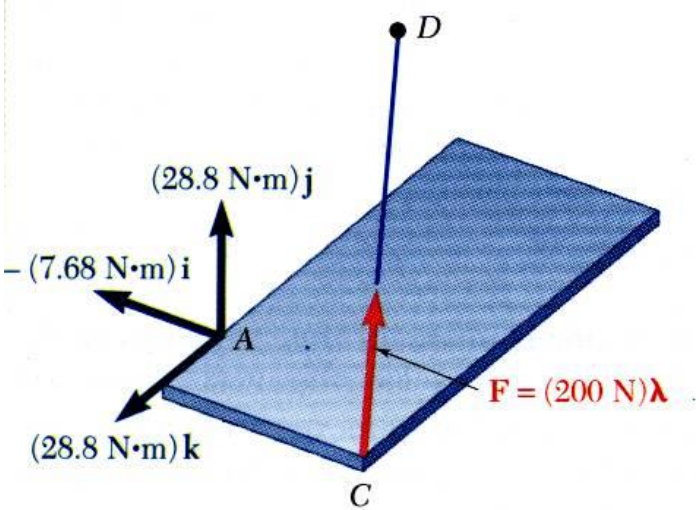
Sample Problem 3.4





SOLUTION:

$$\vec{M}_A = \vec{r}_{C/A} \times \vec{F}$$

$$\vec{r}_{C/A} = \vec{r}_C - \vec{r}_A = (0.3\text{ m})\vec{i} + (0.08\text{ m})\vec{k}$$

$$\vec{F} = F\vec{\lambda} = (200\text{ N})\frac{\vec{r}_{C/D}}{r_{C/D}}$$

$$= (200\text{ N})\frac{-(0.3\text{ m})\vec{i} + (0.24\text{ m})\vec{j} - (0.32\text{ m})\vec{k}}{0.5\text{ m}}$$

$$= -(120\text{ N})\vec{i} + (96\text{ N})\vec{j} - (128\text{ N})\vec{k}$$

$$\vec{M}_A = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0.3 & 0 & 0.08 \\ -120 & 96 & -128 \end{vmatrix}$$

$$\boxed{\vec{M}_A = -(7.68\text{ N}\cdot\text{m})\vec{i} + (28.8\text{ N}\cdot\text{m})\vec{j} + (28.8\text{ N}\cdot\text{m})\vec{k}}$$

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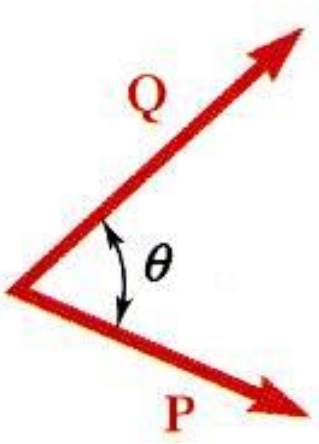
Scalar Product of Two Vectors

- The *scalar product* or *dot product* between two vectors ***P*** and ***Q*** is defined as

$$\vec{P} \bullet \vec{Q} = PQ\cos\theta \quad (\text{scalar result})$$



- Scalar products:
  - are commutative, $\vec{P} \bullet \vec{Q} = \vec{Q} \bullet \vec{P}$
  - are distributive, $\vec{P} \bullet \left(\vec{Q}_1 + \vec{Q}_2\right) = \vec{P} \bullet \vec{Q}_1 + \vec{P} \bullet \vec{Q}_2$
  - are not associative, $\left(\vec{P} \bullet \vec{Q}\right) \bullet \vec{S} = \text{undefined}$

- Scalar products with Cartesian unit components,

$$\vec{P} \bullet \vec{Q} = \left(P_x \vec{i} + P_y \vec{j} + P_z \vec{k}\right) \bullet \left(Q_x \vec{i} + Q_y \vec{j} + Q_z \vec{k}\right)$$

$$\vec{i} \bullet \vec{i} = 1 \quad \vec{j} \bullet \vec{j} = 1 \quad \vec{k} \bullet \vec{k} = 1 \quad \vec{i} \bullet \vec{j} = 0 \quad \vec{j} \bullet \vec{k} = 0 \quad \vec{k} \bullet \vec{i} = 0$$

$$\vec{P} \bullet \vec{Q} = P_x Q_x + P_y Q_y + P_z Q_z$$

$$\vec{P} \bullet \vec{P} = P_x^2 + P_y^2 + P_z^2 = P^2$$

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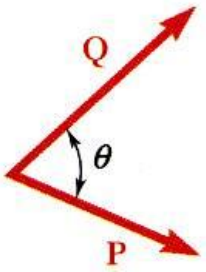
Scalar Product of Two Vectors: Applications

- Angle between two vectors:

$$\vec{P} \bullet \vec{Q} = PQ\cos\theta = P_x Q_x + P_y Q_y + P_z Q_z$$

$$\cos\theta = \frac{P_x Q_x + P_y Q_y + P_z Q_z}{PQ}$$

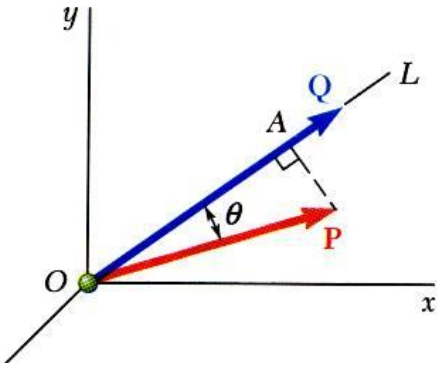


- Projection of a vector on a given axis:

$$P_{OL} = P\cos\theta = \text{projection of } P \text{ along } OL$$

$$\vec{P} \bullet \vec{Q} = PQ\cos\theta$$

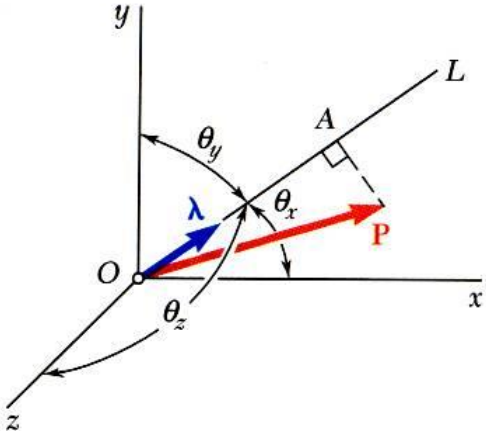
$$\frac{\vec{P} \bullet \vec{Q}}{Q} = P\cos\theta = P_{OL}$$



- For an axis defined by a unit vector:

$$P_{OL} = \vec{P} \bullet \vec{\lambda}$$

$$= P_x \cos\theta_x + P_y \cos\theta_y + P_z \cos\theta_z$$



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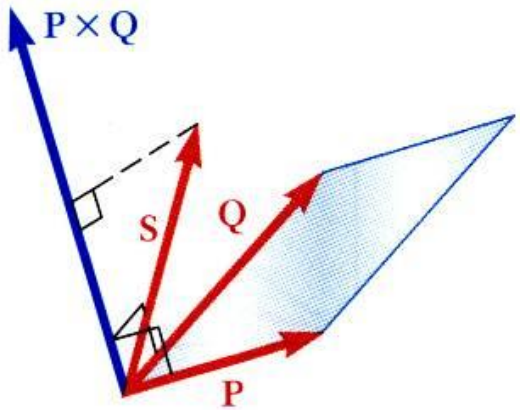
Mixed Triple Product of Three Vectors



- Mixed triple product of three vectors,

$$\vec{S} \bullet \left(\vec{P} \times \vec{Q}\right) = \text{scalar result}$$

- The six mixed triple products formed from ***S***, ***P***, and ***Q*** have equal magnitudes but not the same sign,

$$\vec{S} \bullet \left(\vec{P} \times \vec{Q}\right) = \vec{P} \bullet \left(\vec{Q} \times \vec{S}\right) = \vec{Q} \bullet \left(\vec{S} \times \vec{P}\right)$$
$$= -\vec{S} \bullet \left(\vec{Q} \times P\right) = -\vec{P} \bullet \left(\vec{S} \times \vec{Q}\right) = -\vec{Q} \bullet \left(\vec{P} \times \vec{S}\right)$$

- Evaluating the mixed triple product,

$$\vec{S} \bullet \left(\vec{P} \times \vec{Q}\right) = S_x\left(P_y Q_z - P_z Q_y\right) + S_y\left(P_z Q_x - P_x Q_z\right)$$
$$+ S_z\left(P_{xy} Q_z - P_y Q_x\right)$$
$$= \begin{vmatrix} S_x & S_y & S_z \\ P_x & P_y & P_z \\ Q_x & Q_y & Q_z \end{vmatrix}$$

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Moment of a Force About a Given Axis

- Moment $\boldsymbol{M_O}$ of a force $\boldsymbol{F}$ applied at the point $\boldsymbol{A}$ about a point $\boldsymbol{O}$,

$$\vec{M}_O = \vec{r} \times \vec{F}$$

- Scalar moment M_{OL} about an axis $\boldsymbol{OL}$ is the projection of the moment vector $\boldsymbol{M_O}$ onto the axis,

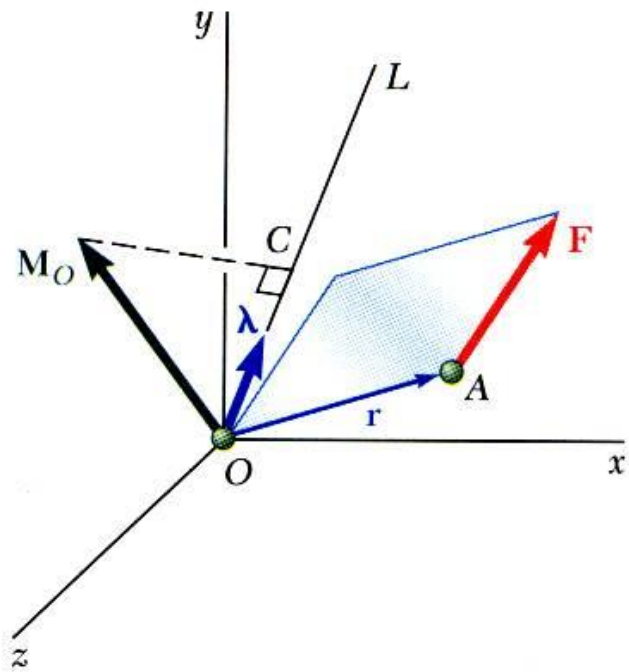
$$M_{OL} = \vec{\lambda} \bullet \vec{M}_O = \vec{\lambda} \bullet \left(\vec{r} \times \vec{F}\right)$$

- Moments of $\boldsymbol{F}$ about the coordinate axes,

$$M_x = yF_z - zF_y$$

$$M_y = zF_x - xF_z$$

$$M_z = xF_y - yF_x$$



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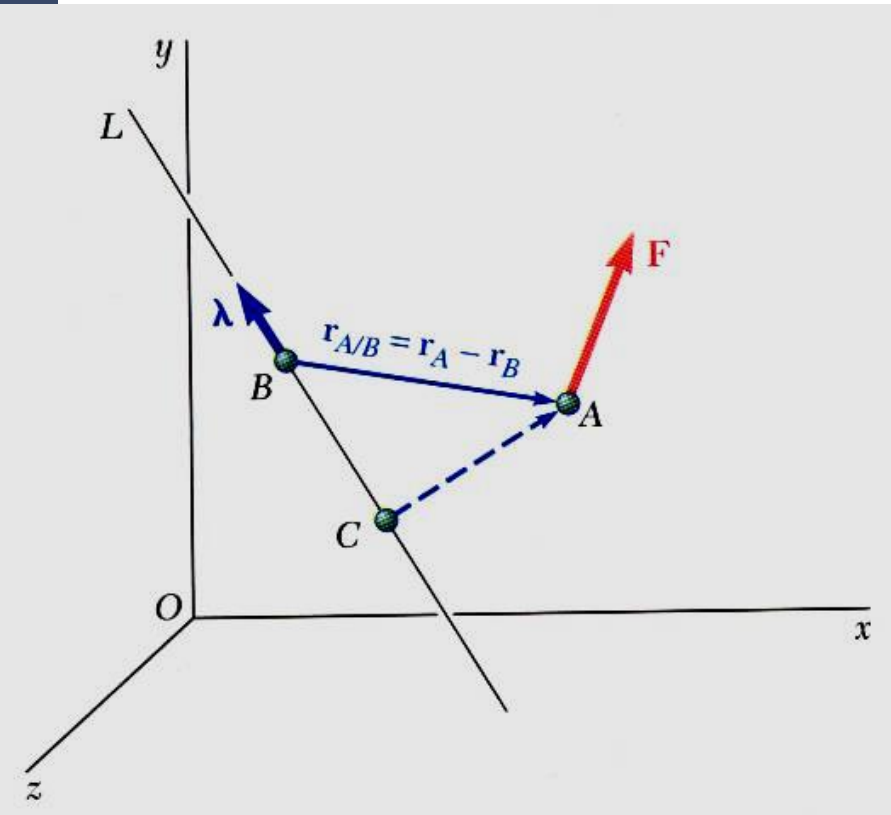
Moment of a Force About a Given Axis



- Moment of a force about an arbitrary axis,

$$M_{BL} = \vec{\lambda} \bullet \vec{M}_B$$
$$= \vec{\lambda} \bullet \left(\vec{r}_{A/B} \times \vec{F}\right)$$
$$\vec{r}_{A/B} = \vec{r}_A - \vec{r}_B$$

- The result is independent of the point *B* along the given axis.

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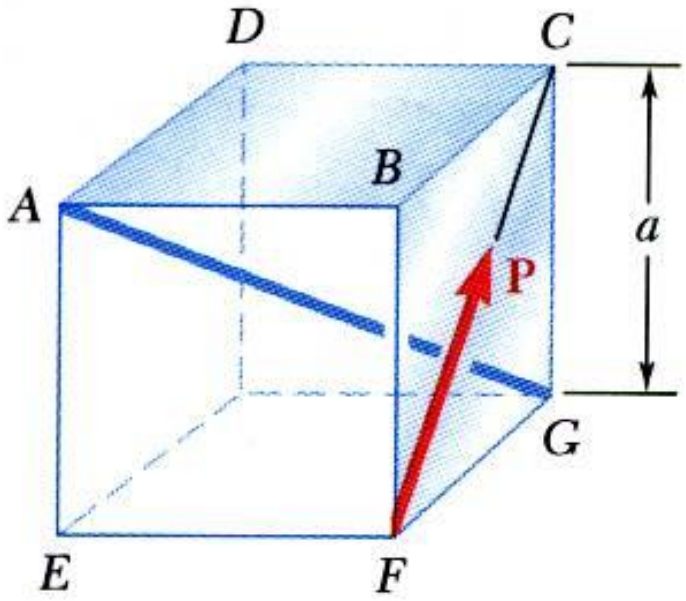
Sample Problem 3.5



A cube is acted on by a force ***P*** as shown. Determine the moment of ***P***

a) about *A*

b) about the edge *AB* and

c) about the diagonal *AG* of the cube.

d) Determine the perpendicular distance between *AG* and *FC*.

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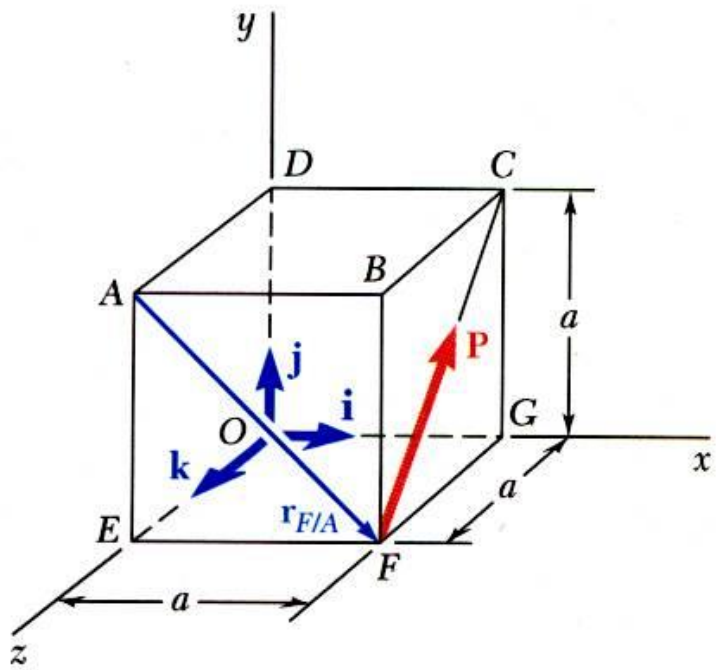
Sample Problem 3.5



- Moment of ***P*** about *A*,

$$\vec{M}_A = \vec{r}_{F/A} \times \vec{P}$$

$$\vec{r}_{F/A} = a\vec{i} - a\vec{j} = a\left(\vec{i} - \vec{j}\right)$$

$$\vec{P} = P/\sqrt{2}\left(\vec{j} - \vec{k}\right)$$

$$\vec{M}_A = a\left(\vec{i} - \vec{j}\right) \times P/\sqrt{2}\left(\vec{j} - \vec{k}\right)$$

$$\boxed{\vec{M}_A = \left(aP\sqrt{2}\right)\left(\vec{i} + \vec{j} + \vec{k}\right)}$$

- Moment of ***P*** about *AB*,

$$M_{AB} = \vec{i} \bullet \vec{M}_A$$

$$= \vec{i} \bullet \left(aP\sqrt{2}\right)\left(\vec{i} + \vec{j} + \vec{k}\right)$$

$$\boxed{M_{AB} = aP\sqrt{2}}$$

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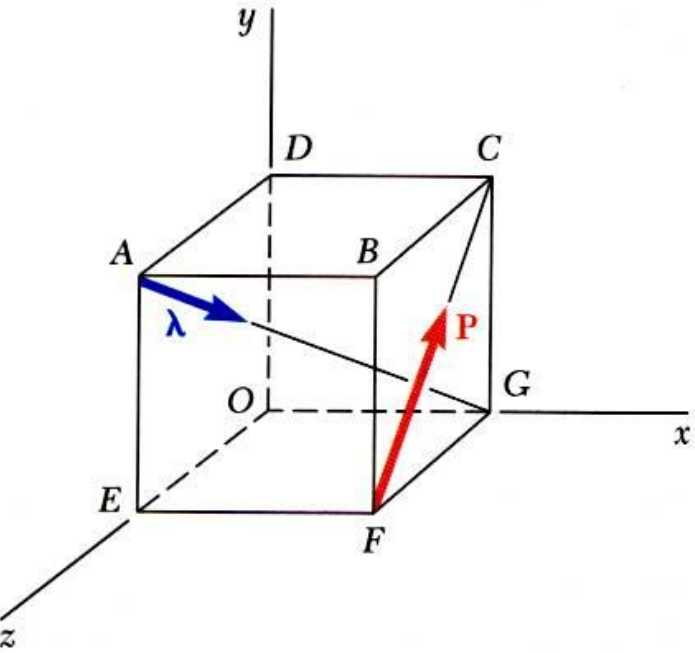
Sample Problem 3.5



- Moment of ***P*** about the diagonal *AG*,

$$M_{AG} = \vec{\lambda} \bullet \vec{M}_A$$

$$\vec{\lambda} = \frac{\vec{r}_{A/G}}{r_{A/G}} = \frac{a\vec{i} - a\vec{j} - a\vec{k}}{a\sqrt{3}} = \frac{1}{\sqrt{3}}\left(\vec{i} - \vec{j} - \vec{k}\right)$$

$$\vec{M}_A = \frac{aP}{\sqrt{2}}\left(\vec{i} + \vec{j} + \vec{k}\right)$$

$$M_{AG} = \frac{1}{\sqrt{3}}\left(\vec{i} - \vec{j} - \vec{k}\right) \bullet \frac{aP}{\sqrt{2}}\left(\vec{i} + \vec{j} + \vec{k}\right)$$

$$= \frac{aP}{\sqrt{6}}(1 - 1 - 1)$$

$$\boxed{M_{AG} = -\frac{aP}{\sqrt{6}}}$$

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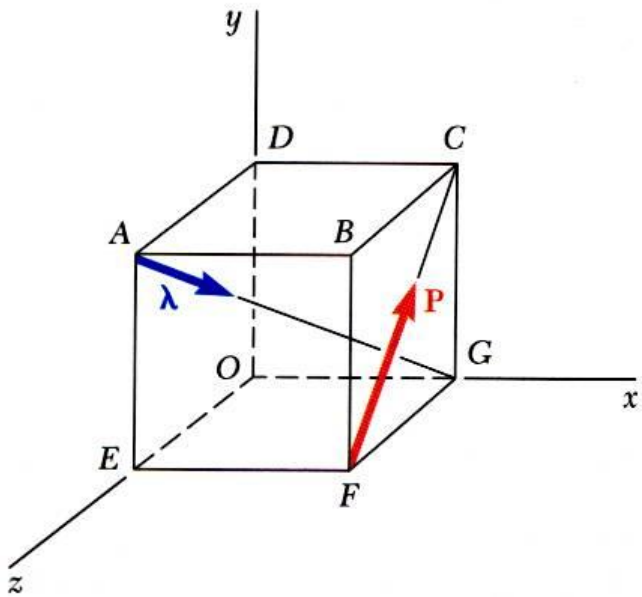
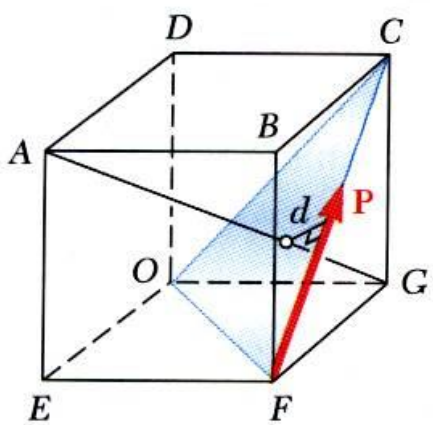
Sample Problem 3.5





- Perpendicular distance between *AG* and *FC*,

$$\vec{P} \bullet \vec{\lambda} = \frac{P}{\sqrt{2}}\left(\vec{j} - \vec{k}\right) \bullet \frac{1}{\sqrt{3}}\left(\vec{i} - \vec{j} - \vec{k}\right) = \frac{P}{\sqrt{6}}(0 - 1 + 1)$$

$$= 0$$

Therefore, *P* is perpendicular to *AG*.

$$\left|M_{AG}\right| = \frac{aP}{\sqrt{6}} = Pd$$

$$\boxed{d = \frac{a}{\sqrt{6}}}$$

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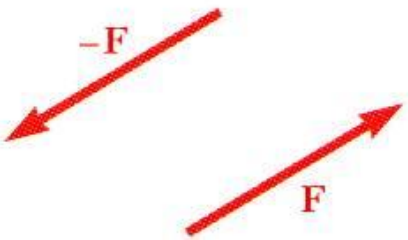
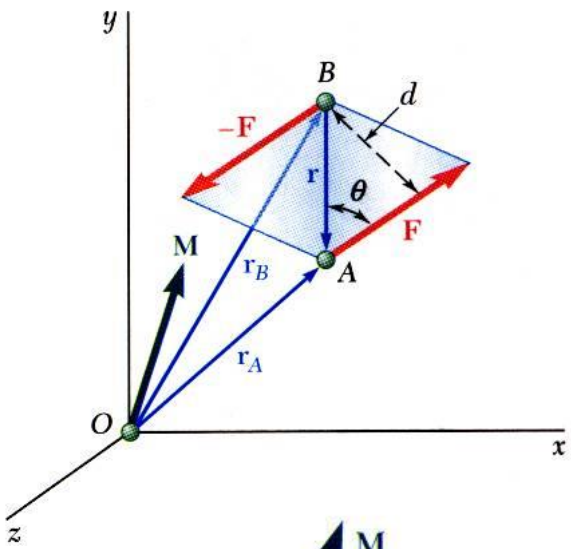
Moment of a Couple

- Two forces ***F*** and -***F*** having the same magnitude, parallel lines of action, and opposite sense are said to form a *couple*.



- Moment of the couple,

$$\vec{M} = \vec{r}_A \times \vec{F} + \vec{r}_B \times \left(-\vec{F}\right)$$
$$= \left(\vec{r}_A - \vec{r}_B\right) \times \vec{F}$$
$$= \vec{r} \times \vec{F}$$
$$M = rF \sin\theta = Fd$$



- The moment vector of the couple is independent of the choice of the origin of the coordinate axes, i.e., it is a *free vector* that can be applied at any point with the same effect.

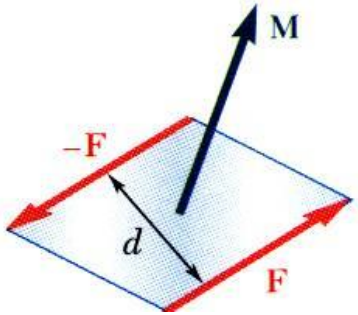


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Photo 3.1 The parallel upward and downward forces of equal magnitude exerted on the arms of the lug nut wrench are an example of a couple.

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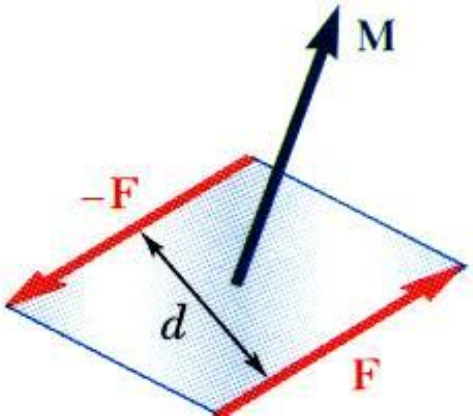
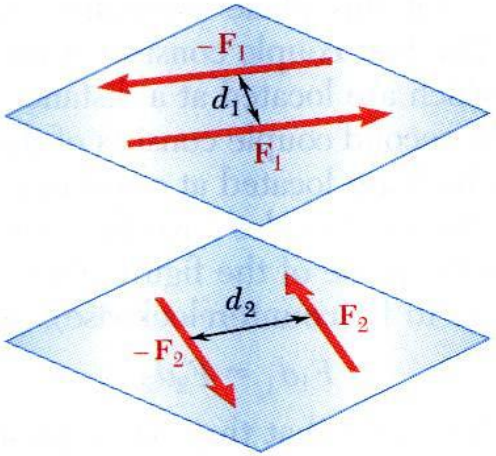
Moment of a Couple

Two couples will have equal moments if

- $F_1 d_1 = F_2 d_2$
- the two couples lie in parallel planes, and
- the two couples have the same sense or the tendency to cause rotation in the same direction.





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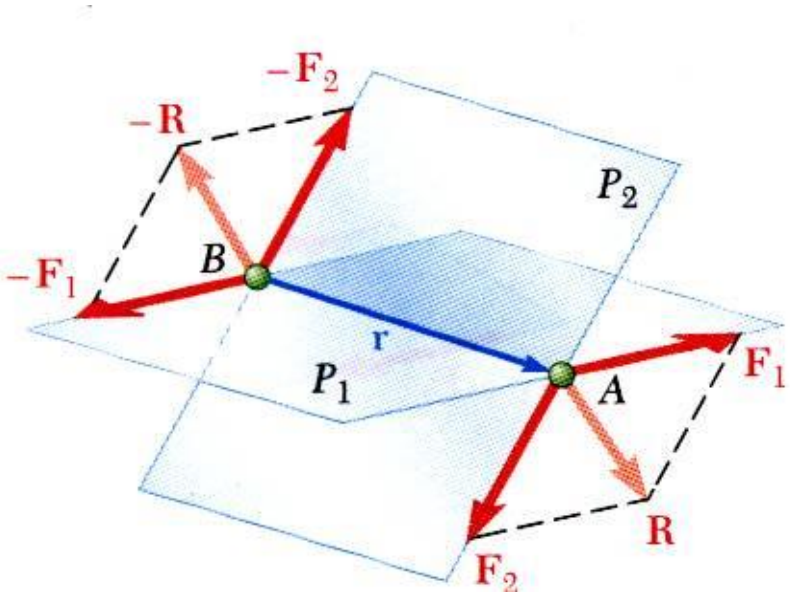
Addition of Couples

- Consider two intersecting planes P_1 and P_2 with each containing a couple

$$\vec{M}_1 = \vec{r} \times \vec{F}_1 \text{ in plane } P_1$$

$$\vec{M}_2 = \vec{r} \times \vec{F}_2 \text{ in plane } P_2$$



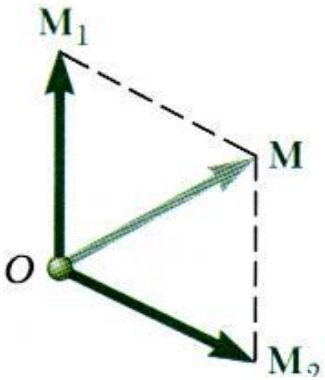
- Resultants of the vectors also form a couple

$$\vec{M} = \vec{r} \times \vec{R} = \vec{r} \times \left(\vec{F}_1 + \vec{F}_2\right)$$

- By Varigon's theorem

$$\vec{M} = \vec{r} \times \vec{F}_1 + \vec{r} \times \vec{F}_2$$

$$= \vec{M}_1 + \vec{M}_2$$



- Sum of two couples is also a couple that is equal to the vector sum of the two couples

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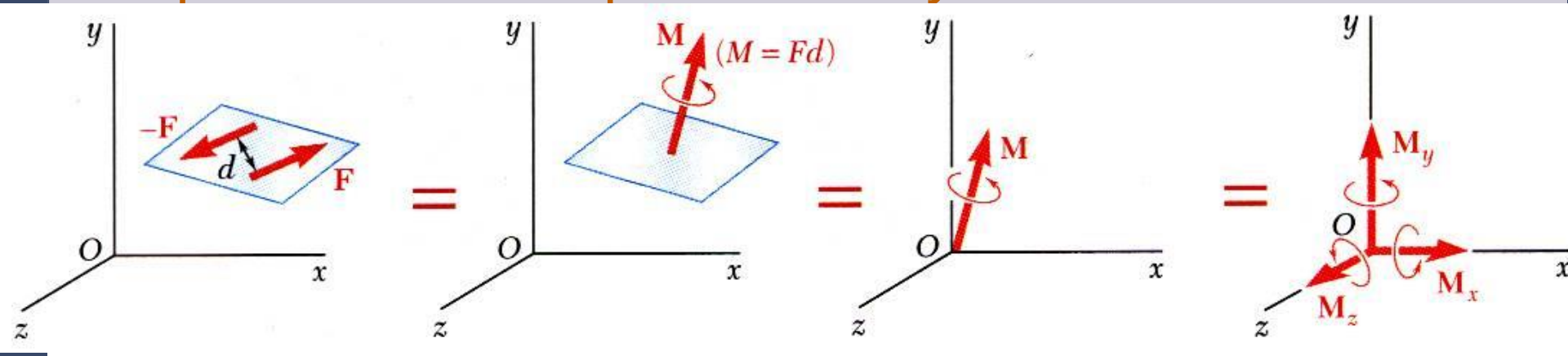
Couples Can Be Represented by Vectors



- A couple can be represented by a vector with magnitude and direction equal to the moment of the couple.
- *Couple vectors* obey the law of addition of vectors.
- Couple vectors are free vectors, i.e., the point of application is not significant.
- Couple vectors may be resolved into component vectors.

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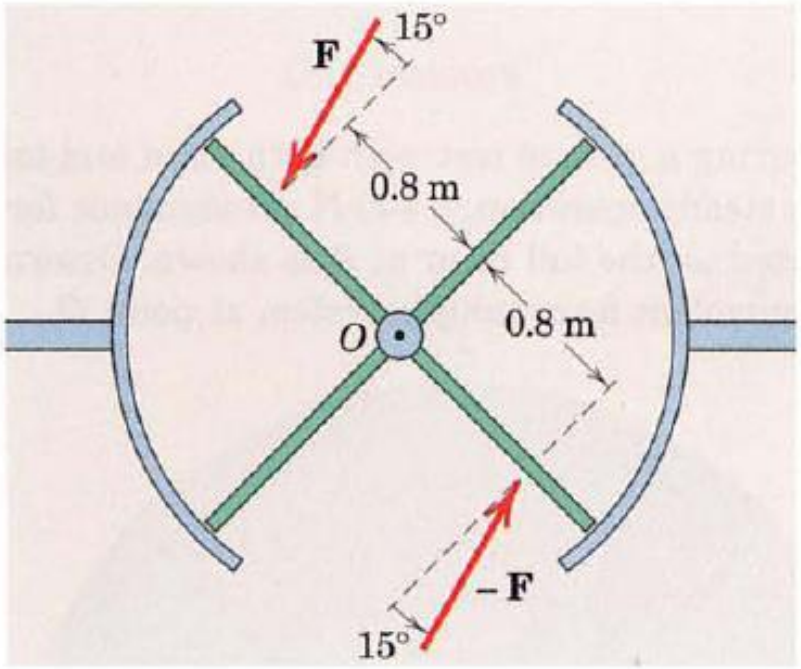
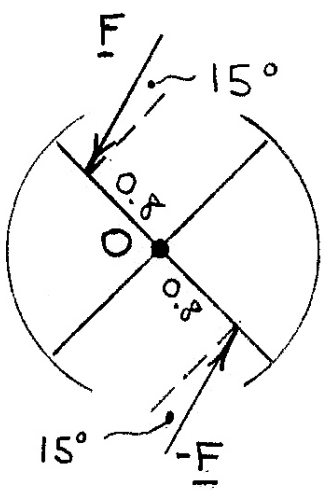
Problem

The top view of a revolving entrance door is shown. Two persons simultaneously approach the door and exert forces of equal magnitudes as shown. If the resulting moment about the door pivot axis at O is 25 N·m, determine the force magnitude F.



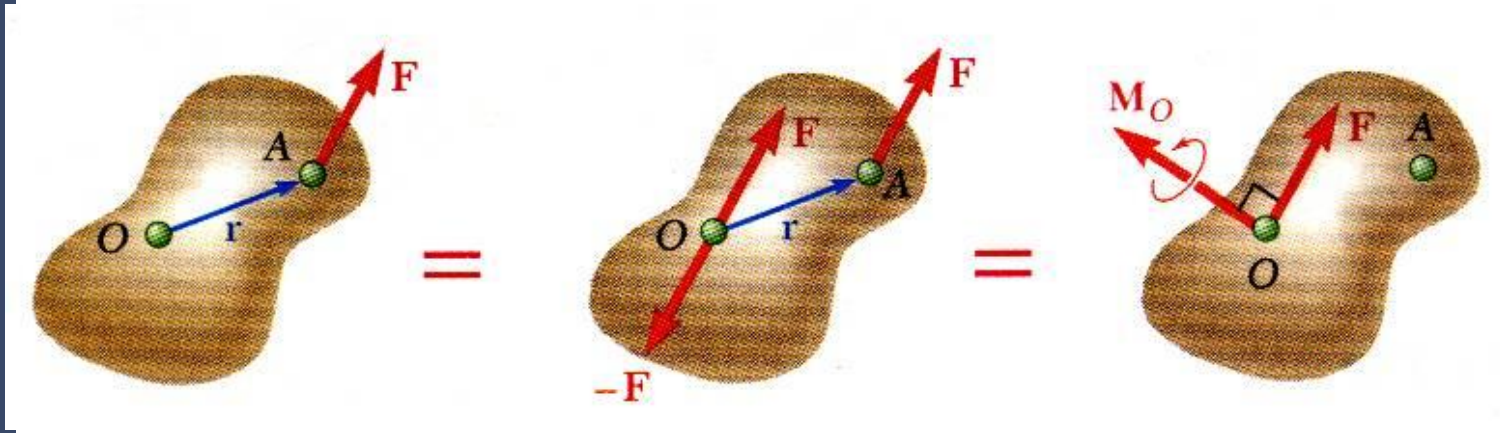


$$\circlearrowleft + \; M_o = \Sigma F d$$

$$25 = 2F(\cos 15°)(0.8)$$

$$F = \underline{16.18 \text{ N}}$$

Resolution of a Force Into a Force at *O* and a Couple



- Force vector ***F*** can not be simply moved to *O* without modifying its action on the body.
- Attaching equal and opposite force vectors at *O* produces no net effect on the body.
- The three forces may be replaced by an equivalent force vector and couple vector, i.e, a *force-couple system*.

The force exerted by each hand on the wrench could be replaced with an equivalent force-couple system acting on the nut.

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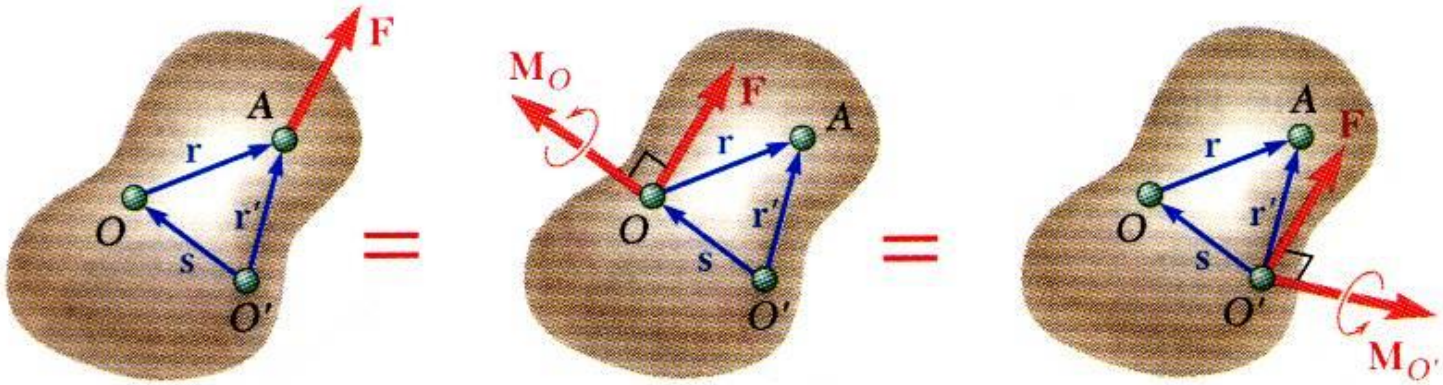
Resolution of a Force Into a Force at *O* and a Couple



- Moving ***F*** from *A* to a different point *O'* requires the addition of a different couple vector $\boldsymbol{M_{O'}}$,

$$\vec{M}_{O'} = \vec{r}' \times \vec{F}$$

- The moments of ***F*** about O and *O'* are related,

$$\vec{M}_{O'} = \vec{r}' \times \vec{F} = (\vec{r} + \vec{s}) \times \vec{F} = \vec{r} \times \vec{F} + \vec{s} \times \vec{F}$$

$$= \vec{M}_O + \vec{s} \times \vec{F}$$

- Moving the force-couple system from *O* to *O'* requires the addition of the moment of the force at *O* about *O'*.

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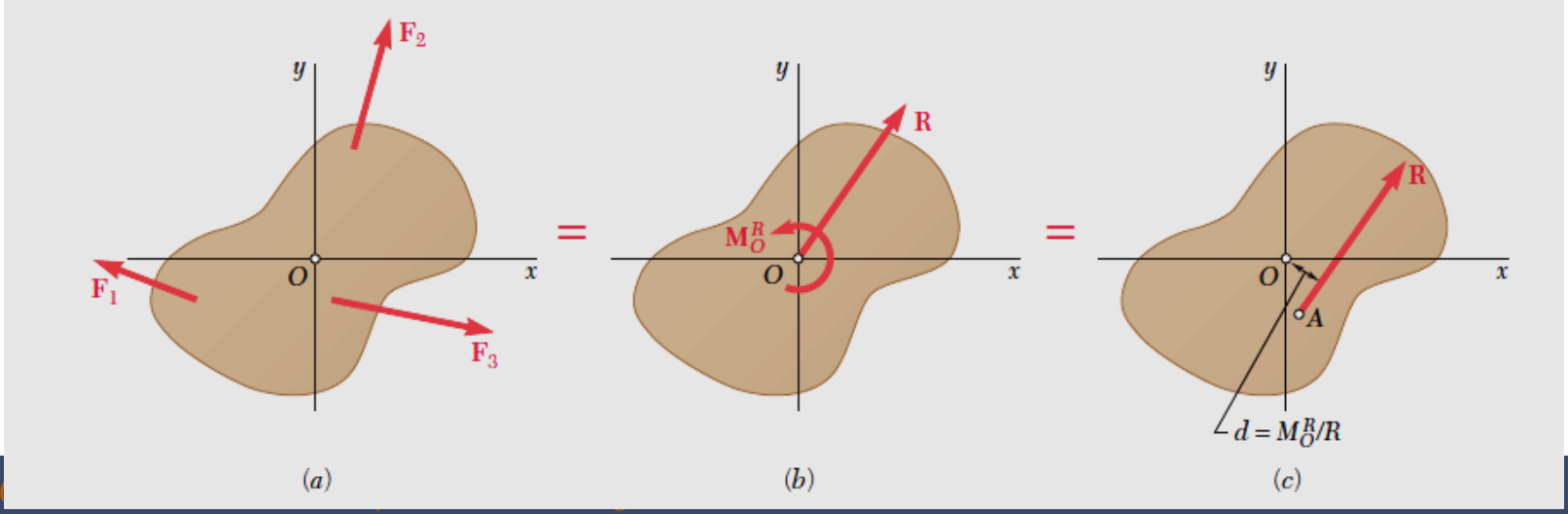
FURTHER REDUCTION OF A SYSTEM OF FORCES

1. ***Concurrent forces are applied at the same point and can therefore*** be added directly to obtain their resultant R . Thus, they always reduce to a single force.

2. ***Coplanar forces act in the same plane, which may be assumed*** *to* be the plane of the figure (Fig. *a*).

   The sum R of the forces of the system will also lie in the plane of the figure, while the moment of each force about *O*, *and thus the moment resultant* $\mathbf{M_O^R}$, *will be perpendicular to that plane. The force-couple system* at *O consists, therefore, of a force R and a couple vector* M_O^R which are mutually perpendicular (Fig. *b*). *They can be* reduced to a single force R by moving R in the plane of the figure until its moment about *O becomes equal to* M_O^R. *The distance* from *O to the line of action of R is d* = M_O^R/R.

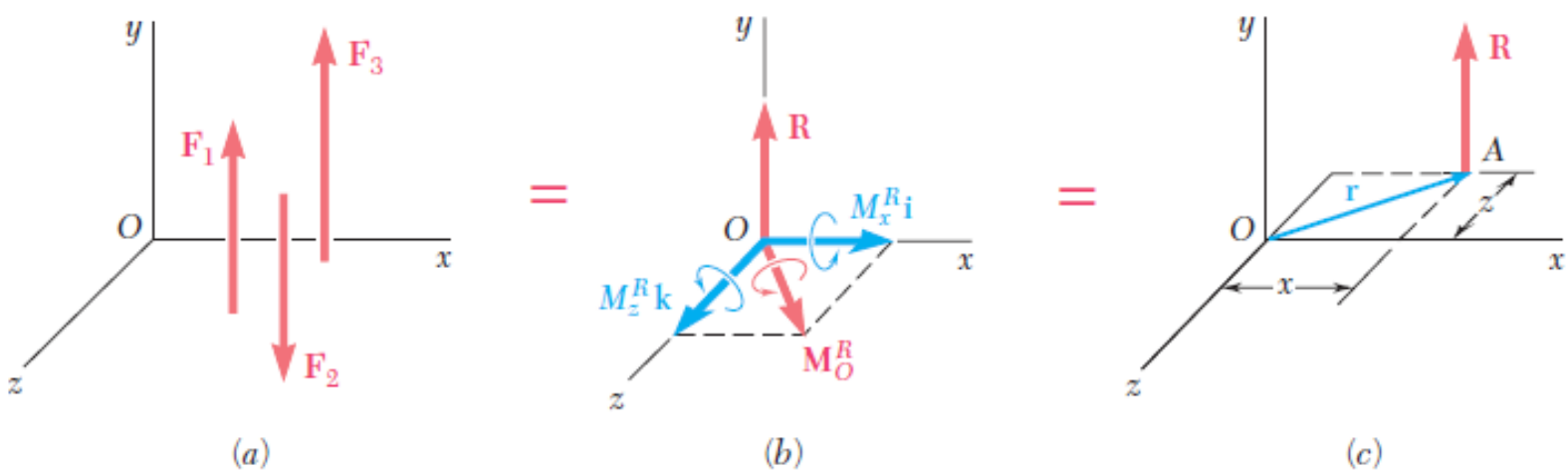


FURTHER REDUCTION OF A SYSTEM OF FORCES

3. ***Parallel forces have parallel lines of action and may or may not*** have the same sense.

Assuming here that the forces are parallel to the *y axis (Fig. a), we note that their sum* ***R will also be parallel*** to the *y axis. On the other hand, since the moment of a given* force must be perpendicular to that force, the moment about *O of* each force of the system, and thus the moment resultant M_O^R***, will*** lie in the *zx plane*.

The force-couple system at O consists, therefore, of a force **R and a couple vector** M_O^R ***which are mutually perpendicular*** (Fig. *b). They can be reduced to a single force* ***R*** (Fig. *c) or, if* ***R = 0, to a single couple of moment*** M_O^R**.**



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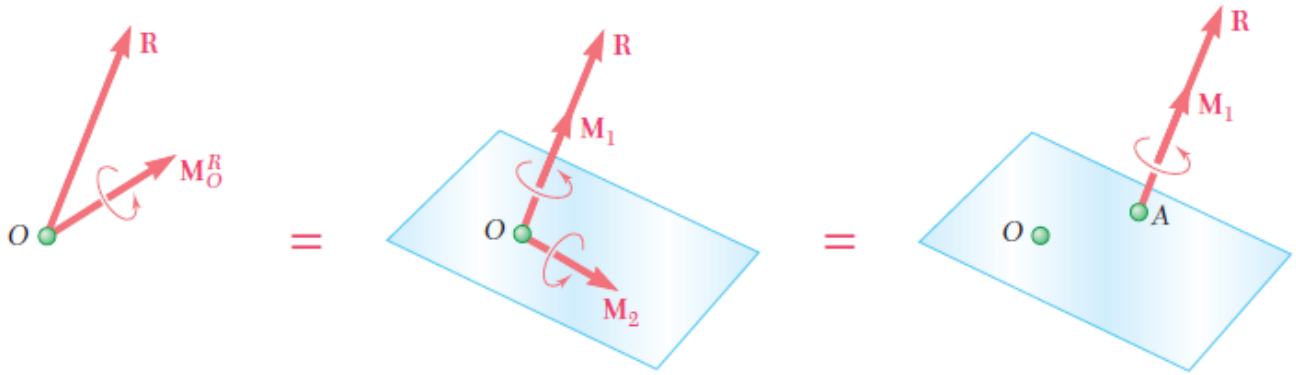
REDUCTION OF A SYSTEM OF FORCES TO A WRENCH



- In the general case of a system of forces in space, the equivalent force couple system at *O consists of a force* ***R and a couple vector*** M_O^R ***which*** are not perpendicular, and neither of which is zero.

- Thus, the system of forces *cannot be reduced to a single force or to a single* couple.

- The couple vector, however, can be replaced by two other couple vectors obtained by resolving M_O^R *into a component* M_1 *along* R and a component M_2 in a plane perpendicular to R (Fig. *b*).

- The couple vector M_2 and the force R can then be replaced by a single force R acting along a new line of action. The original system of forces thus reduces to R and to the couple vector M_1 (Fig. *c*), i.e., to R and a couple acting in the plane perpendicular to R .

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Wrench

- This particular force-couple system is called a *wrench because the resulting* combination of push and twist is the same as that which would be caused by an actual wrench.

- A wrench, therefore, consists of two collinear vectors, namely, a force **R and a couple vector**

$$\mathbf{M}_1 = p\boldsymbol{R}$$

The pushing-turning action associated with the tightening of a screw illustrates the collinear lines of action of the force and couple vector that constitute a wrench.

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Photo 3.4 The parallel wind forces acting on the highway signs can be reduced to a single equivalent force. Determining this force can simplify the calculation of the forces acting on the supports of the frame to which the signs are attached.

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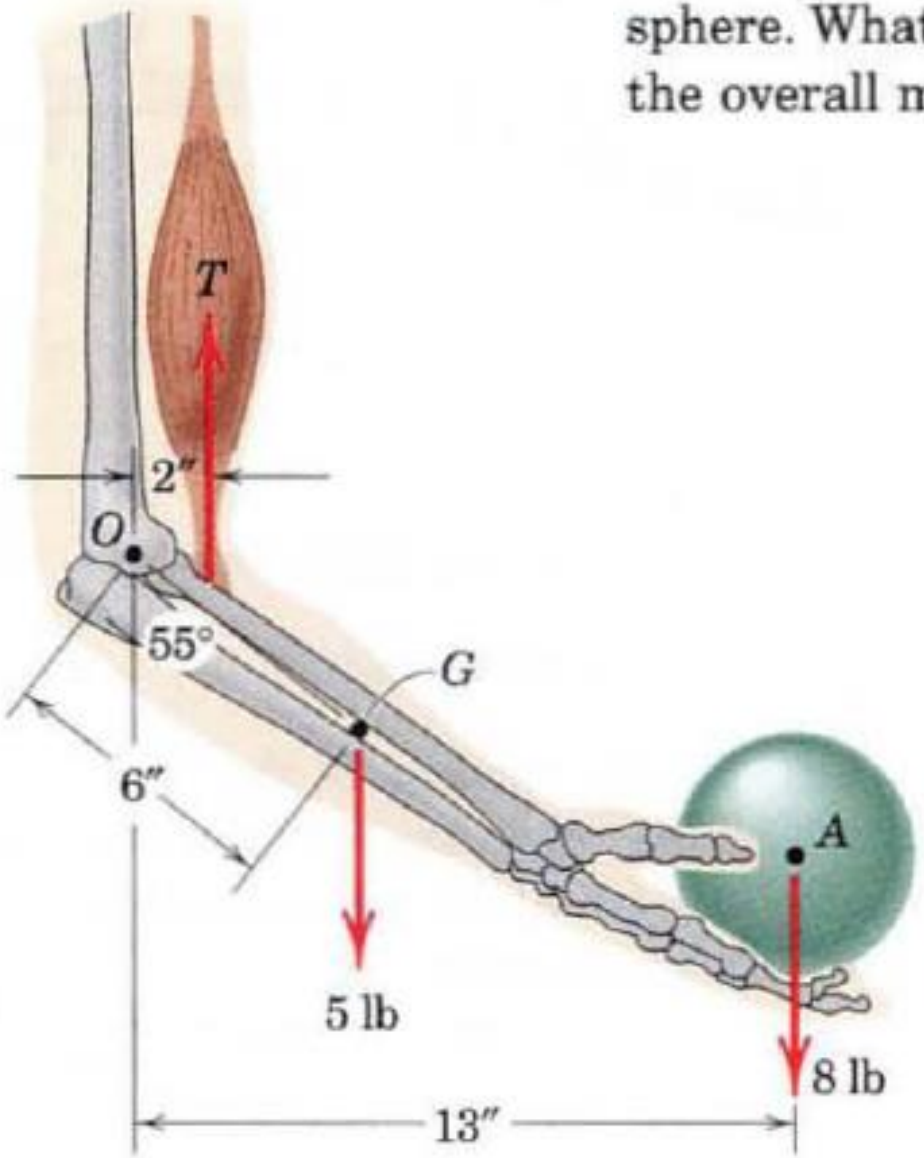
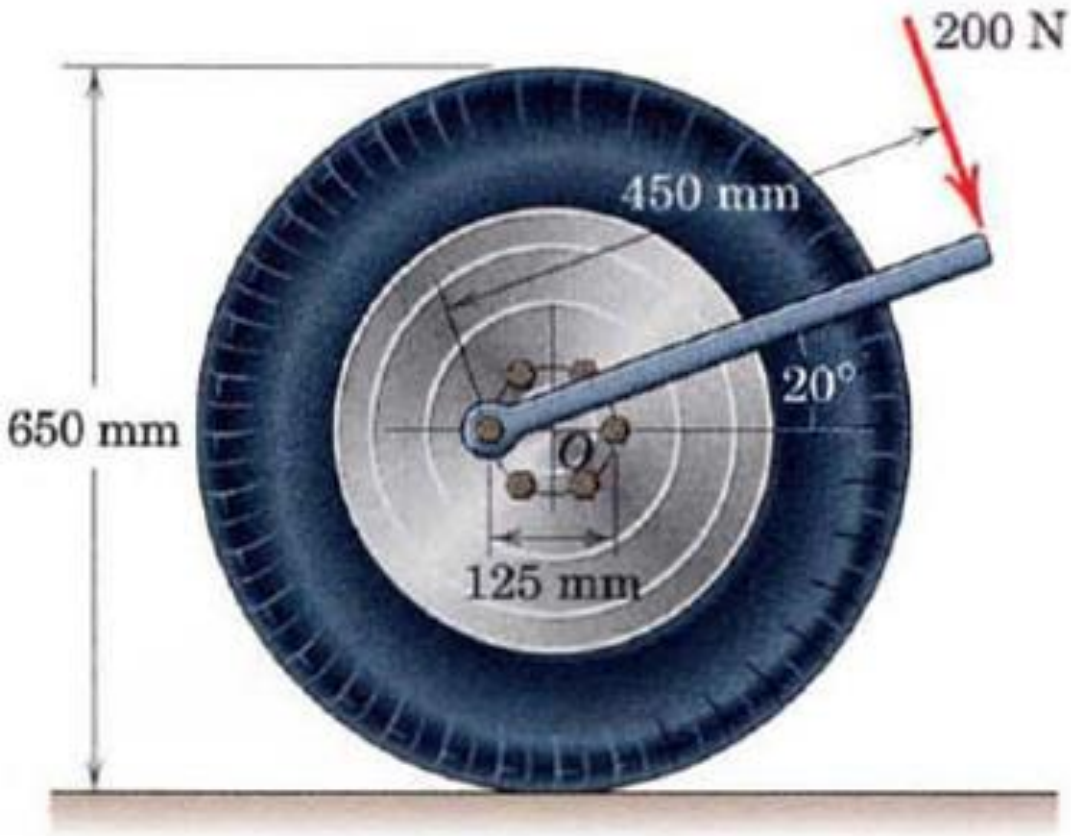
Example Prob.

Elements of the lower arm are shown in the figure. The weight of the forearm is 5 lb with mass center at G. Determine the combined moment about the elbow pivot O of the weights of the forearm and the sphere. What must the biceps tension force be so that the overall moment about O is zero?



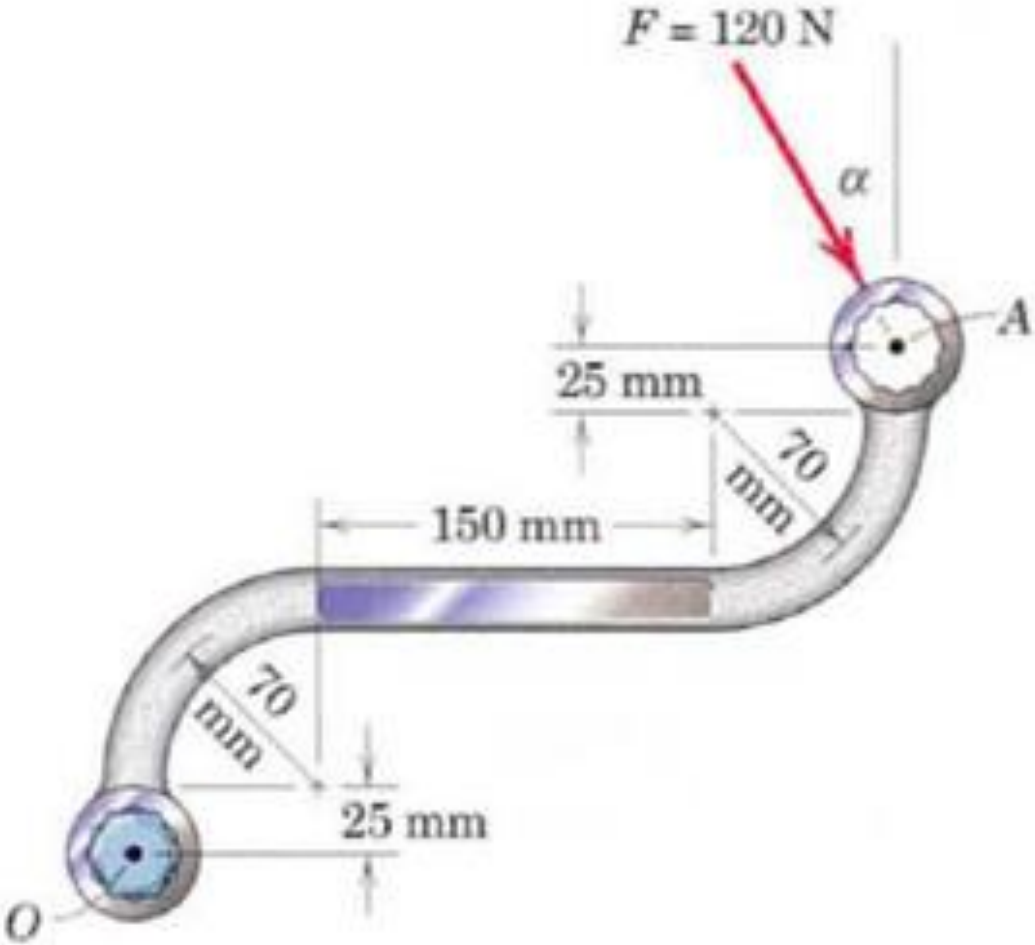
Example Prob.

A force of 200 N is applied to the end of the wrench to tighten a flange bolt which holds the wheel to the axle. Determine the moment M produced by this force about the center O of the wheel for the position of the wrench shown.



Exa

The 120-N force is applied as shown to one end of the curved wrench. If $\alpha = 30°$, calculate the moment of F about the center O of the bolt. Determine the value of α which would maximize the moment about O: state the value of this maximum moment.

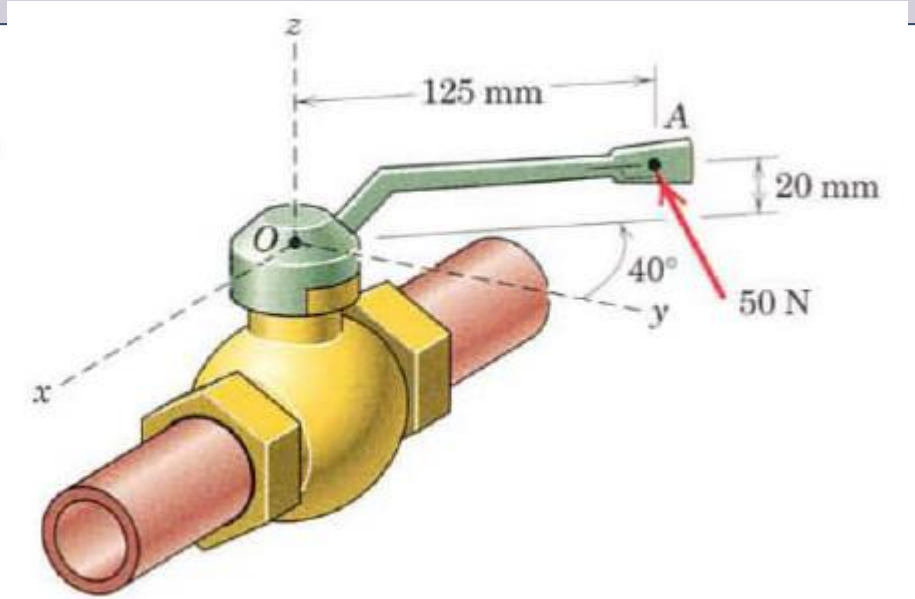


Problem

A 50-N horizontal force is applied to the handle of the industrial water valve as shown. The force is perpendicular to the vertical plane containing line OA of the handle. Determine the equivalent force–couple system at point O.

Ans. $\mathbf{R} = -38.3\mathbf{i} - 32.1\mathbf{j}$ N

$\mathbf{M}_O = 643\mathbf{i} - 766\mathbf{j} + 6250\mathbf{k}$ N·mm



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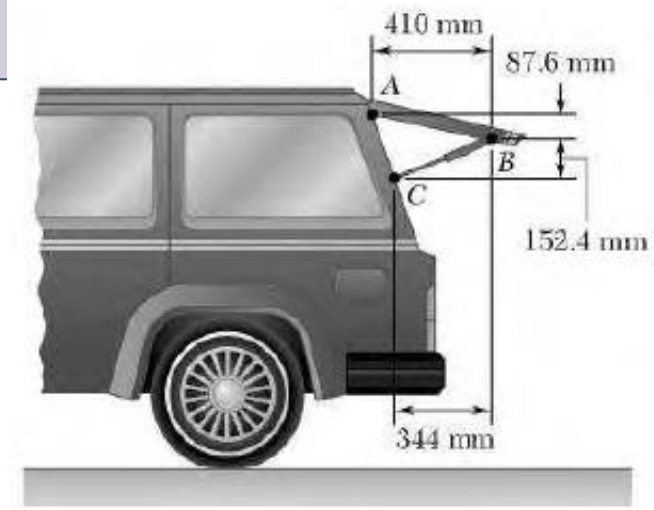
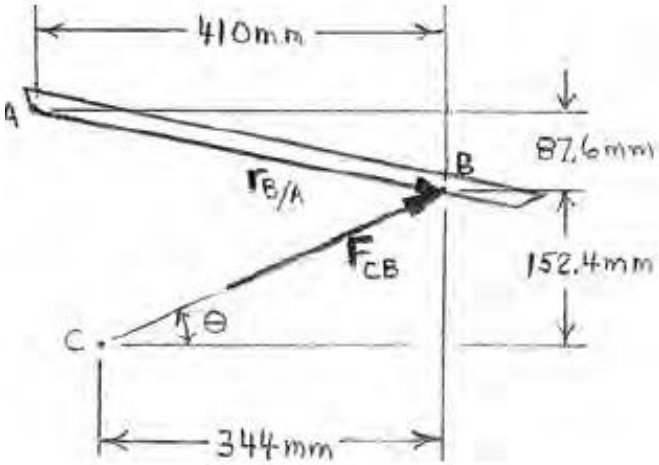
Problem

The tailgate of a car is supported by the hydraulic lift BC. If the lift exerts a 125-N force directed along its center line on the ball and socket at B, determine the moment of the force about A.





$$\mathbf{M}_A = \mathbf{r}_{B/A} \times \mathbf{F}_{CB}$$

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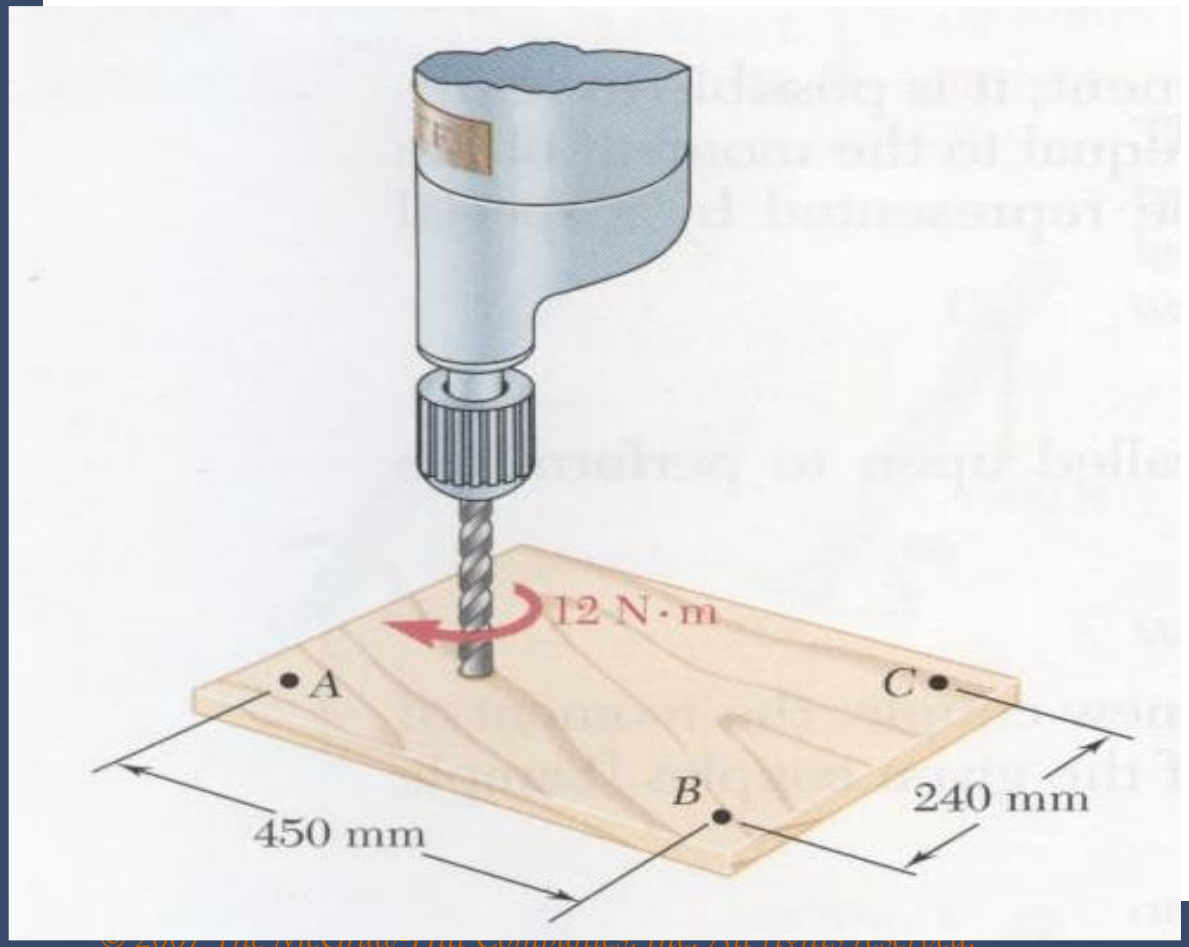
A piece of plywood in which several holes are being drilled successively has been secured to a workbench by means of two nails. Knowing that the drill exerts a 12-N · m couple on the piece of plywood, determine the magnitude of the resulting forces applied to the nails if they are located (*a*) at *A* and *B*, (*b*) at *B* and *C*, (*c*) at *A* and *C*.



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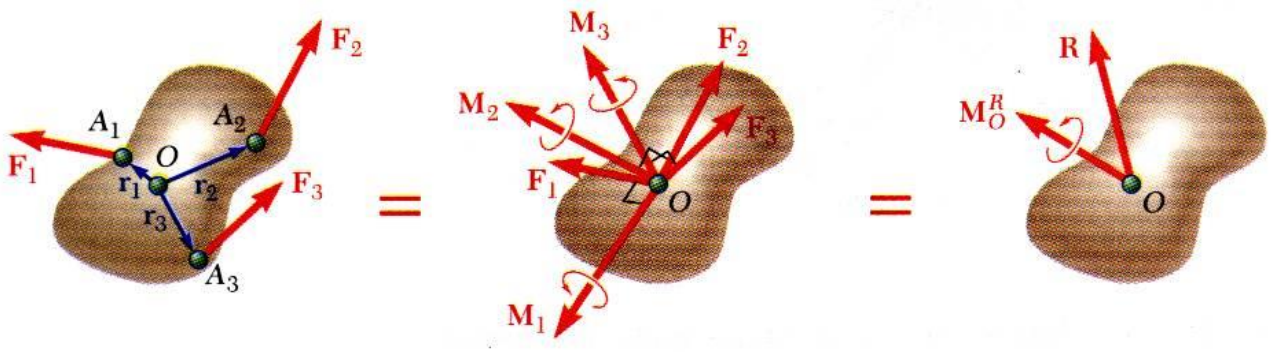
System of Forces: Reduction to a Force and Couple



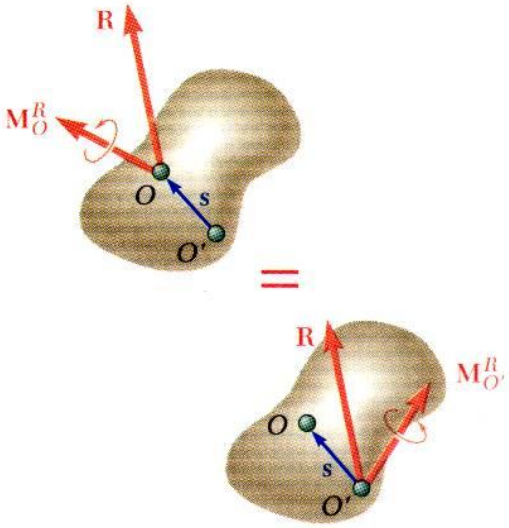
- A system of forces may be replaced by a collection of force-couple systems acting a given point *O*
- The force and couple vectors may be combined into a resultant force vector and a resultant couple vector,

$$\vec{R} = \sum \vec{F} \qquad \vec{M}_O^R = \sum \left(\vec{r} \times \vec{F}\right)$$

- The force-couple system at *O* may be moved to *O'* with the addition of the moment of ***R*** about *O'* ,

$$\vec{M}_{O'}^R = \vec{M}_O^R + \vec{s} \times \vec{R}$$

- Two systems of forces are equivalent if they can be reduced to the same force-couple system.



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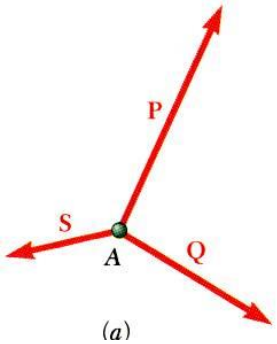
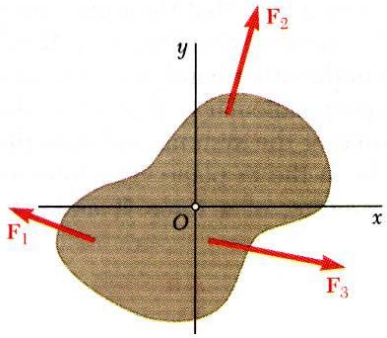
Further Reduction of a System of Forces

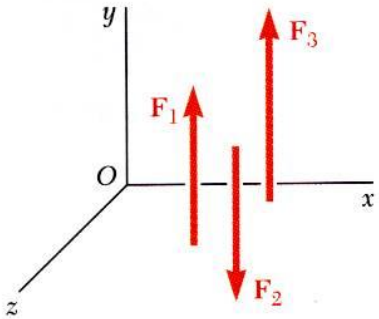
- If the resultant force and couple at *O* are mutually perpendicular, they can be replaced by a single force acting along a new line of action.

- The resultant force-couple system for a system of forces will be mutually perpendicular if:
  1) the forces are concurrent,
  2) the forces are coplanar, or
  3) the forces are parallel.



(a)





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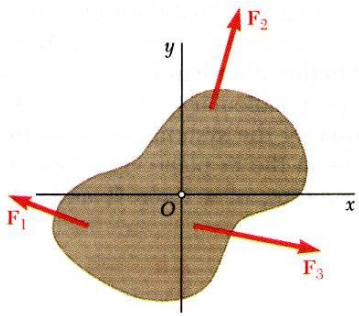
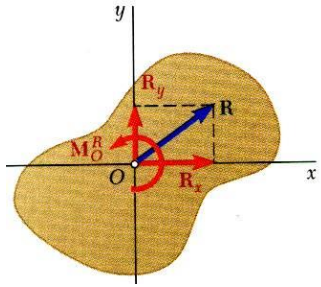
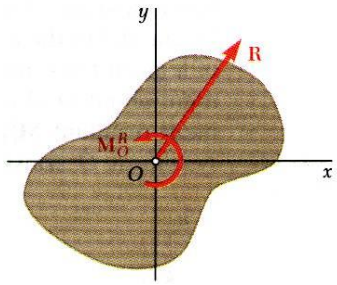
Further Reduction of a System of Forces

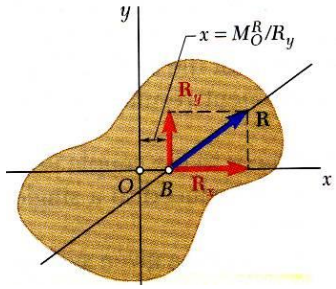


- System of coplanar forces is reduced to a force-couple system $\vec{R}$ and $\vec{M}_O^R$ that is mutually perpendicular.



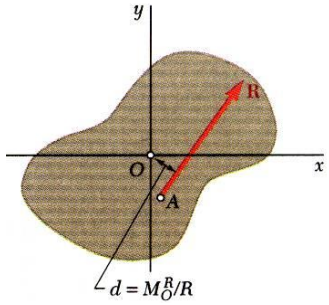


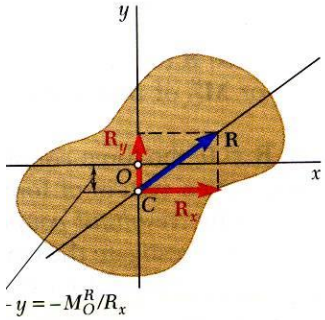
- System can be reduced to a single force by moving the line of action of $\vec{R}$ until its moment about O becomes $\vec{M}_O^R$



- In terms of rectangular coordinates,

$$xR_y - yR_x = M_O^R$$





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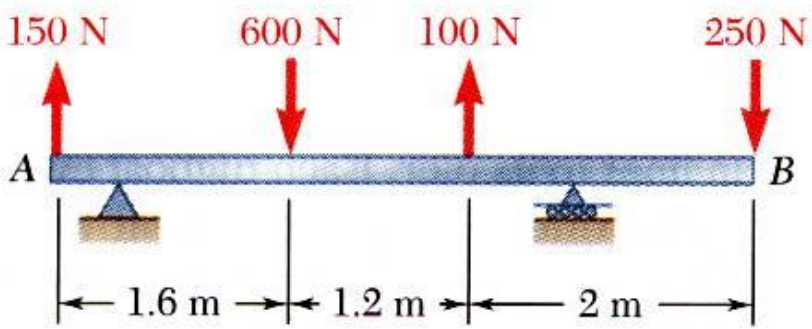
Sample Problem 3.8



For the beam, reduce the system of forces shown to (a) an equivalent force-couple system at *A*, (b) an equivalent force couple system at *B*, and (c) a single force or resultant.

Note: Since the support reactions are not included, the given system will not maintain the beam in equilibrium.

SOLUTION:

a) Compute the resultant force for the forces shown and the resultant couple for the moments of the forces about *A*.

b) Find an equivalent force-couple system at *B* based on the force-couple system at *A*.

c) Determine the point of application for the resultant force such that its moment about *A* is equal to the resultant couple at *A*.

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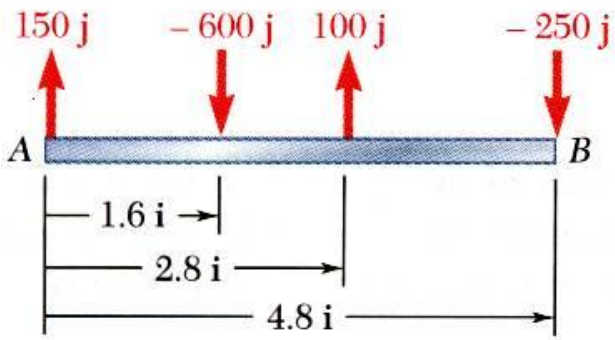
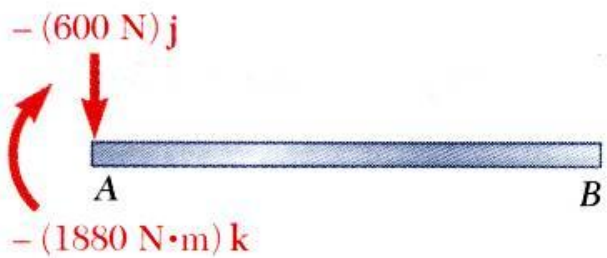
Sample Problem 3.8





SOLUTION:

a) Compute the resultant force and the resultant couple at *A*.

$$\vec{R} = \sum \vec{F}$$

$$= (150\text{ N})\vec{j} - (600\text{ N})\vec{j} + (100\text{ N})\vec{j} - (250\text{ N})\vec{j}$$

$$\boxed{\vec{R} = -(600\text{ N})\vec{j}}$$

$$\vec{M}_A^R = \sum (\vec{r} \times \vec{F})$$

$$= (1.6\vec{i}) \times (-600\vec{j}) + (2.8\vec{i}) \times (100\vec{j})$$

$$+ (4.8\vec{i}) \times (-250\vec{j})$$

$$\boxed{\vec{M}_A^R = -(1880\text{ N}\cdot\text{m})\vec{k}}$$

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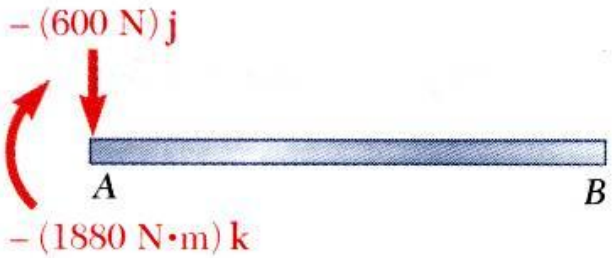
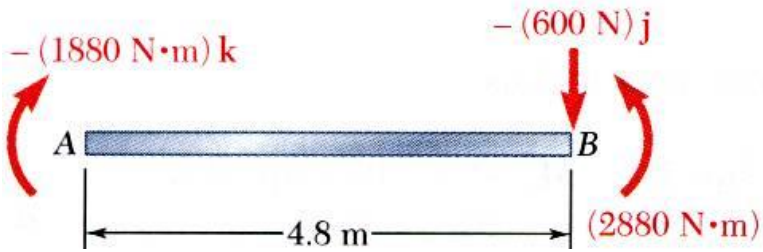
Sample Problem 3.8



b) Find an equivalent force-couple system at *B* based on the force-couple system at *A*.

The force is unchanged by the movement of the force-couple system from *A* to *B*.

$$\boxed{\vec{R} = -(600\text{ N})\vec{j}}$$

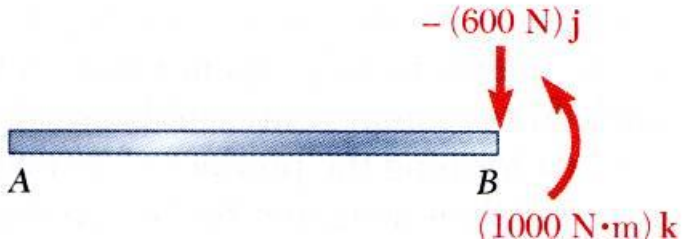


The couple at *B* is equal to the moment about *B* of the force-couple system found at *A*.

$$\vec{M}_B^R = \vec{M}_A^R + \vec{r}_{B/A} \times \vec{R}$$

$$= -(1880\text{ N}\cdot\text{m})\vec{k} + (-4.8\text{ m})\vec{i} \times (-600\text{ N})\vec{j}$$

$$= -(1880\text{ N}\cdot\text{m})\vec{k} + (2880\text{ N}\cdot\text{m})\vec{k}$$



$$\boxed{\vec{M}_B^R = +(1000\text{ N}\cdot\text{m})\vec{k}}$$

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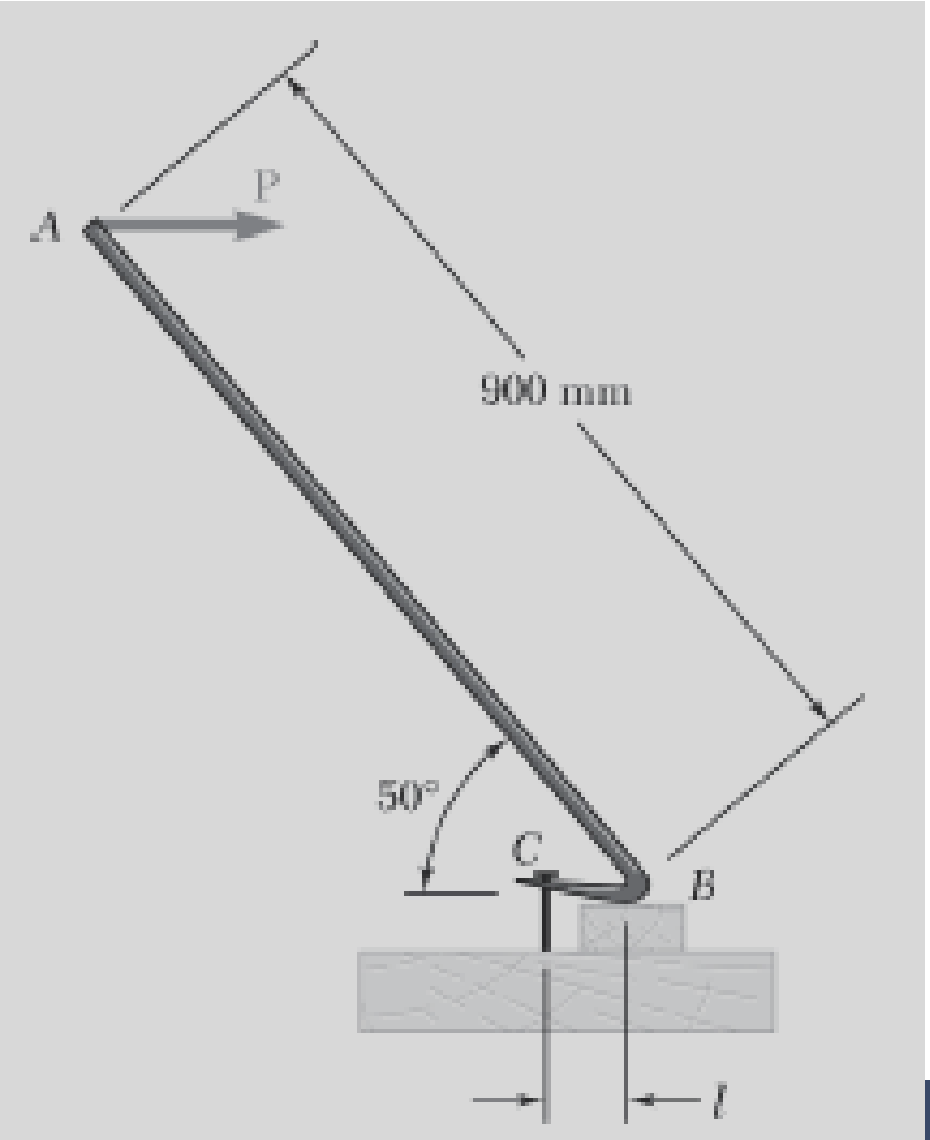
P

To remove a nail, a small block of wood is placed under a crowbar, and a horizontal force **P** is applied as shown. Knowing that $l = 88$ mm and $P = 130$ N, determine the vertical force exerted on the nail and the reaction at B.



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